

Solar-Charged Electric Vehicles: A Comprehensive Analysis of Grid, Driver, and Environmental Benefits

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Abstract—To date, solar-powered electric vehicles (EVs) have often been considered as niche projects or with small vehicle rooftop panels that can slightly extend the electric driving range. This article proposes a large-scale solar EV concept with low-cost, flexible, and thin-film solar cells integrated onto the steel of all upward-facing vehicle body panels as a viable solution to help mitigate EV charging and range concerns and the high cost and solar power intermittency of individual residential rooftop solar installations. Only by considering the full range of benefits, namely to the grid, driver, and environment, the value of solar-charged EVs can truly be appreciated. This article models the effect of panel tilt and partial shading on the solar energy capture of 150 drivers to analyze grid, driver, and environmental benefits in Los Angeles (LA) and Detroit over the course of a full year. The simulations predict net annual vehicle energy use reductions of 21.5% in LA and 17.5% in Detroit for average cloud conditions, compared to a nonsolar EV.

Index Terms—Battery chargers, modeling, power transformer thermal factors, solar energy, vehicles.

I. INTRODUCTION

ELECTRIC vehicles (EVs) have the potential to significantly reduce greenhouse gas emissions, regulated emissions which cause local air pollution and negative health effects [1], and society's reliance on fossil fuels. Although EV adoption has been slow in the past, the pace has recently been increasing: 2.1 million plug-in vehicles were sold globally in 2018, a 64% increase from 2017 [2]. Still, in 2018, plug-in vehicles accounted for only 2.2% of the global light vehicle market [2]. Continuing challenges for EV adoption include cost, charging concerns, and range concerns. As EV adoption continues to increase, the generation mix and transmission limitations for charging EVs will become a pressing matter.

With regard to electricity generation, it is well known that the full environmental benefits of EVs can only be realized when charging energy is generated with 0 or low CO₂ emissions (e.g., wind, solar, and hydro) and that the use of electricity generated from coal will negate most of the CO₂ benefits of EVs compared to conventional internal combustion engine (ICE) vehicles. Thus, renewable energy

generation and increased EV adoption must go hand-in-hand. Current photovoltaic (PV) cell technology provides an energy conversion efficiency of 26.7% for the highest efficiency silicon monocrystalline cells [3], with record 1-sun efficiencies of 38.8% for laboratory demonstrations of multijunction cells [4] (which are not economically practical to date). Most notably for EVs, the recent advances have created flexible thin-film cells with efficiencies in excess of 20% [5]. Solar deployment has been increasing worldwide, particularly in utility-scale deployments; however, its impact at the individual level (residential rooftop solar) has been slow due to the large upfront cost for installation (e.g., \$23 000 for a 7.5-kW system) [6]. Furthermore, its intermittency (daily and seasonally) and variability (due to weather conditions) make it an auxiliary source. Its intermittency can be addressed with energy storage, which adds an additional significant cost and is a further barrier to deployment for the average homeowner. Thus, many EV owners rely on charging from their local grid, and common evening EV charging adds demand to the grid at peak load times, exacerbating grid emissions.

With regard to electricity transmission and distribution for EV charging, the losses and load limitations must be considered. There is an average of 12% loss in energy from large generating stations to local residential loads [7], so for EV charging, it means that the EV is actually consuming 12% more generated energy than the energy used to drive the vehicle. Furthermore, most of the grids present today were not designed for EV charging. Much research has investigated this issue, and a primary concern of having an increasing number of EVs plugging into the grid to charge is the overloading of distribution transformers [8]–[12]. Most distribution transformers presently in use have been sized to provide the peak loads of the connected homes without EV charging in mind, and thus, the additional loading from EV charging can cause overheating, accelerated aging, and eventual transformer failure. Some previous research has investigated the effect of EV charging on distribution transformers with the use of residential rooftop PV generation: ElNozahy and Salama [13] and Uriarte *et al.* [14] investigated transformer overloading levels only, Geiles and Islam [15] investigated the associated transformer loss-of-life in Australia, and Abdelsamad *et al.* [16] analyzed transformer loss-of-life under North American loading conditions, finding that at 50% EV penetration, transformer replacement could be deferred by nearly 4 years when residential rooftop PV

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was used. Recently, El-Bataway and Morsi [17] investigated the distribution transformer loss-of-life considering residential rooftop solar shingles, fast chargers, and second-generation battery energy storage. However, no previous work has analyzed distribution transformer overloading and accelerated aging with EVs utilizing onboard solar generation.

To improve both the solar installation cost and storage issues and the EV charging and range issues, this article proposes the concept of an EV with low-cost flexible thin-film PV cells integrated directly onto the steel of all upward-facing body panels of the vehicle (i.e., roof, hood, and trunk). The solar intermittency issue is resolved because the EV has a large built-in battery already, and thus, the required energy storage does not represent an additional cost as it does with residential solar systems. PV integration during the vehicle manufacturing process opens up the possibility for low-cost mass production, seamless integration, and negligible additional mass of the flexible thin-film cells, as opposed to the more expensive and rigid traditional monocrystalline silicon cells. Copper indium gallium selenide (CIGS) cells are currently the best available choice due to their flexibility, proven ability to integrate onto steel [18], and high efficiency: in this article, 20% of nominal efficiency is assumed though 23.35% has been demonstrated [19], [20]. Although the solar power generated will be low compared to the power needed to accelerate a vehicle, this article will show that the energy obtained over a full day parked or driving outside is significant in terms of reduced grid charging needs at the end of the day and for driving range extension. With a target mass production additional cost of \$670 (\$620 for the PV cells [21] and integration and \$50 for power electronics, as discussed in Section IV-D), the proposed solar EV (SEV) option would be affordable for a large segment of the population buying EVs, with mature vehicle financing services making the additional cost much more manageable to the average buyer than a residential rooftop installation.

In the past, SEVs have generally been considered niche development projects, yet some automotive companies are now starting to experiment with solar cells on passenger vehicles. For example, the Toyota Prius Prime and the Karma Revero both have solar cells integrated onto the roof, which charge the propulsion battery [22]. Alta, a maker of thin-film gallium arsenide (GaAs) solar cells, claims solar cell efficiencies around 30% [23] though GaAs cells are currently expensive. Audi plans to integrate these Alta cells on the roof of its 2020 EV line though the solar power will initially only power the low-voltage system and not the propulsion battery [23]. Furthermore, Sono Motors GmbH, Munich, Germany, is currently developing the Sion EV, which has monocrystalline silicon PV modules fixed onto all vehicle body panels to provide electric range extension [24]. However, this article proposes that the significant benefits can be achieved using low-cost thin-film PV cells, even with lower nominal efficiency of 20%. A recent research also shows increasing interest in PV on vehicles: Duan *et al.* [25] proposed EV battery balancing powered by vehicle-rooftop solar panels, Grosso *et al.* [26] developed an electrical architecture for charging the battery from onboard monocrystalline PV cells, Ding *et al.* [27] proposed

reconfigurable PV arrays for on-vehicle use to deal with partial shading, and Abdelhamid *et al.* [28] analyzed the benefits of adding monocrystalline silicon PV cells to ICE vehicles though do not use detailed vehicle models nor real-world driving data. Onboard solar has also been investigated for other modes of transportation: Mulhall *et al.* [29] presented the design of a solar-assisted electric rickshaw, Diab-Marzouk and Trescases [30] presented a silicon carbide-based converter with differential power processing for a solar-powered aircraft, and Gan *et al.* [31] presented a switched reluctance motor drive for a solar-assisted hybrid electric bus. However, no previous work has examined the potential system-level benefits of SEVs with low cost and widespread flexible thin-film PV cell integration.

The proposed full-scale SEV concept using low-cost, lightweight PV cells integrated onto the steel of all upward-facing vehicle body panels makes sense from many other perspectives. First, there is a very direct and efficient path for solar energy from PV cell to EV battery, with fewer power conversions than a stationary PV panel charging a stationary storage battery, which later transfers energy to the EV battery. In terms of EV range extension, the full-scale SEV concept is the ultimate wireless charging solution, using free and clean solar energy to charge the battery when driving or parked anywhere outside. Although much research has been conducted on dynamic wireless charging through the road, obstacles remain including enormous infrastructure investment needed to retrofit roads, and peak driving times often coincide with afternoon peak grid loads [32], [33], meaning that new generation could be needed, local grid generation mix might not guarantee low CO₂ driving, and grid transmission losses (about 12%) still exist. From the perspective of vehicle efficiency, automotive engineers spend much effort to increase EV efficiencies by a few percentages: aerodynamic improvements, more efficient motors, and power electronics, and there are diminishing returns in this area for the current generation of highly efficient EVs. However, the proposed full-scale SEV can reduce charging needs by about 20% on average, depending on driving scenarios, which is analogous to an increase in efficiency of 20%. There is not a single other EV development that has the potential for such large grid energy use reductions.

There are also significant challenges to be overcome before mass-produced SEVs can become a reality. First, a low-cost process must be developed to integrate PV cells seamlessly onto the automotive steel panels, with good durability in harsh weather and over time. Second, the configuration of series and parallel PV cells must be optimized for maximum power point tracking (MPPT) over the curved vehicle surfaces and generating high voltages which are easier to boost to the voltage of the traction battery. Third, MPPT tracking controllers must be designed to quickly respond to fast-changing partial shading patterns which can occur when driving. Fourth, the electrical architecture and associated power electronic converters, which transfer the PV energy to the high-voltage traction battery, must be designed and optimized for high efficiency, low cost, and low mass. Since there is much future work required to solve these challenges, it is imperative to first assess the system-level benefits of the proposed SEV concept, in order

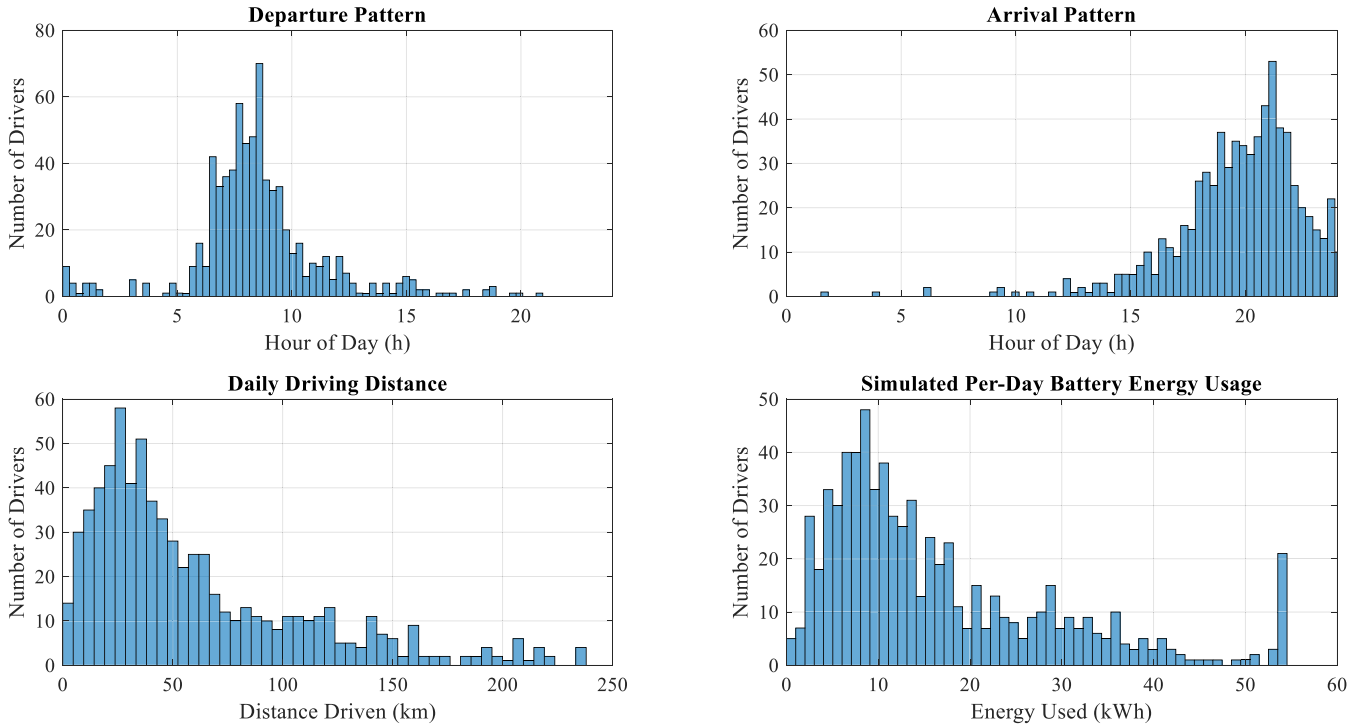


Fig. 1. Distribution of departure and arrival times, daily distance traveled, and simulated daily energy requirement for 150 logged drivers over 5 weekend days.

to determine if the future design and development effort are warranted—this is the goal of this article.

An earlier related work analyzed the transformer aging reductions from ten SEV drivers in 1 summer week [34] and found significant benefits, indicating that further analysis of SEV system-level benefits is merited. However, actual transformer failures and replacements are based on annual equivalent transformer aging, including aging factors that are much lower in cooler months due to the lower ambient temperatures—thus, the summer-only analysis does not accurately represent annual transformer aging. Furthermore, Mobarak *et al.* [34] do not address driver or environmental benefits, do not model the effect of panel tilt, vehicle orientation, or partial shading, nor do it discuss practical SEV challenges. Thus, the contributions of this article are as follows.

- 1) Modeling and analyzing the effect of panel tilt, vehicle orientation, and partial shading on captured solar energy at different latitudes.
- 2) Adding temperature dependence to the efficiency of the solar cells.
- 3) Investigating grid, driver, and environmental benefits of solar-powered EVs (SEVs) using logged driving cycles from 150 drivers.
- 4) Expanding the analysis to an entire year, which yields new results that cannot be obtained from a summer-only analysis.
- 5) Investigating the transformer aging benefit of SEVs in the case of delayed charging.
- 6) Performing a preliminary analysis on the case where extra solar energy is fed back to the home.

- 7) Discussing practical engineering challenges and future trends on the path toward mass-produced SEVs.

Section II describes the data sources used in this research. Section III discusses the vehicle modeling, solar cell modeling and challenges, SEV electrical architecture challenges, and transformer aging modeling. Section IV presents the analysis results for energy use, grid benefits, driver benefits, environmental benefits, and economic analysis, respectively. Section V discusses the future SEV trends, and Section VI provides the conclusion.

II. DATA SOURCES

A. Vehicle Data

This analysis uses 1 week of logged vehicle data from 150 drivers of ICE vehicles in Toronto, ON, Canada. Data loggers were connected to the onboard diagnostic port of each participant's vehicle to log CANbus (controller area network) signals including date, time, and second-by-second vehicle speeds of each trip. Since altitude signals were not available on the CANbus, the road grade of each trip is unknown, and thus, a grade is assumed to be 0. ICE driving data are an excellent data source for EV modeling because the logged drivers have no daily range limitations and can thus drive as they naturally need to. The MATLAB scripts were created to organize the driving days, which were assumed to start at 12 A.M. Figs. 1 and 2 show the driving habits of the logged drivers for the 5 weekdays and 2 weekend days logged—the distribution curves of these data sets are described mathematically in [35]. The logged speed from driving data is fed directly into the SEV Simulink model. The

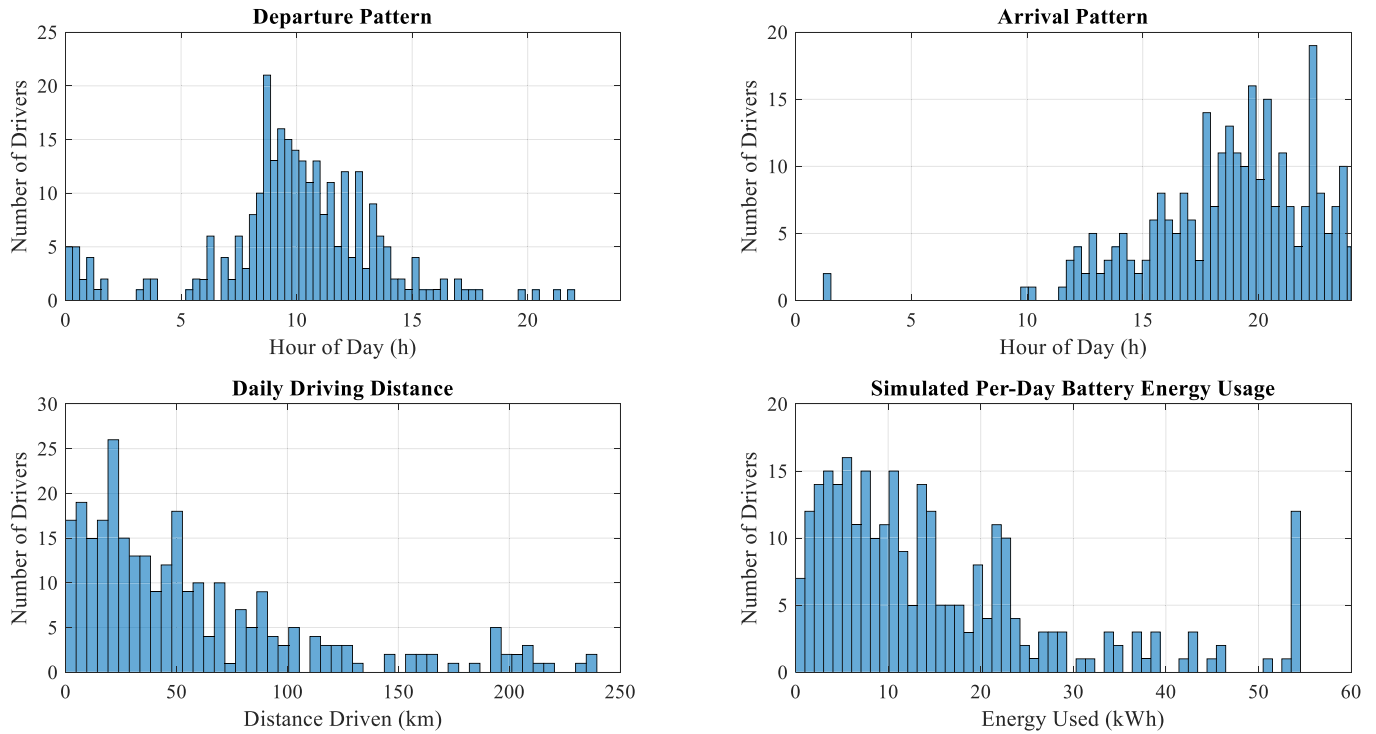


Fig. 2. Distribution of departure and arrival times, daily distance traveled, and simulated daily energy requirement for 150 logged drivers over 2 weekend days.

simulated per-day energy use was based on a vehicle model of a Chevrolet Bolt driving the logged cycles, which will be described in Section III. The simulated battery energy use for this base case assumes a 2-kW accessory power load and a useable battery capacity of 54 kWh, and thus, there is a spike in the histograms at 54-kWh energy use to account for the ICE vehicles that drove further than the Chevrolet Bolt model could drive.

B. Grid Data

The transformer aging analysis relies on the relationship between the quantity and timing of EV charging to the distribution transformer ratings. This article assumes a distribution grid with ten houses per distribution transformer and a maximum of one EV per house. Thus, 15 transformers are considered, meaning at 100% EV penetration (ratios of EVs to homes), and each of the 150 unique drivers is assigned to one house. Thus, in an area that commonly has two cars per home, a 100% EV penetration rate would mean that 50% of the cars on the road are electric. This is considered as the upper limit in this article, which could be approached in certain neighborhood clusters or regions in the mid-to-long-term future. For EV penetration rates less than 100%, EVs are assigned randomly within the group of 150 houses to mimic actual EV uptake within a neighborhood, from 0% to 100% EV penetration, in steps of 10%. Fig. 3 shows the monthly per house average residential load profiles without EV charging, as obtained from [36]. In [36], data for every 2nd month of the year were gathered from 200 houses in Florida and then averaged. In this article, profiles for every month have been generated by averaging the 2 months around

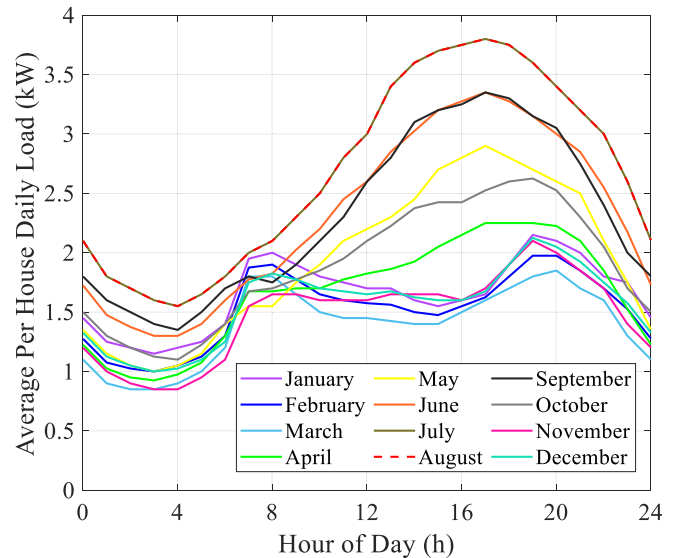


Fig. 3. Monthly per house load profile without EV charging.

the skipped month. The exception is August, where the load profile is set equal to the July profile. Fig. 3 shows an average household load of 2.7 kW and a peak of 3.8 kW in July and an average load of 1.6 kW and a peak of 2.2 kW in January.

Distribution transformer sizes are often chosen based on: 1) yearly aging factor and 2) maximum overloading. The aging factor will vary month to month based on changing loads and the ambient temperature. The transformer is sized so that even with planned overloading, the annual

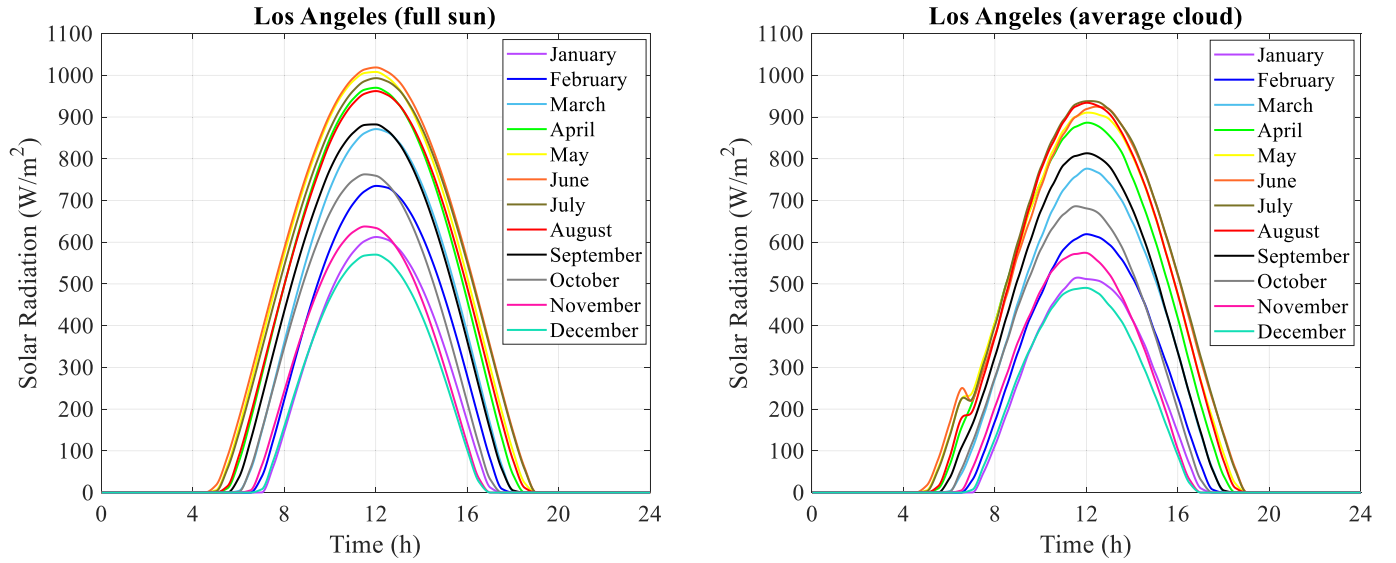


Fig. 4. Monthly solar radiation in LA for sunny and average cloud condition days.

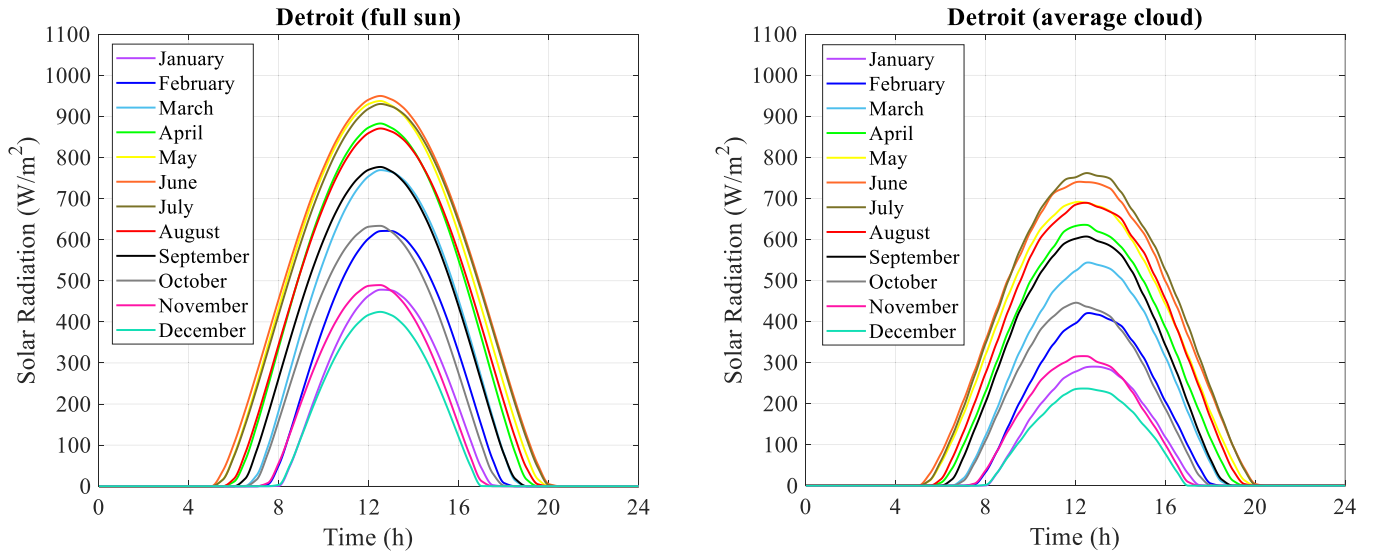


Fig. 5. Monthly solar radiation in Detroit for sunny and average cloud condition days.

equivalent aging factor (F_{EQA}) equals to one, meaning that the transformer has aged 1 year over 1 year of use. Second, maximum transformer overloading must not exceed 200% for power transformers and 300% for distribution transformers [37]. For the given grid load profile, a 25-kVA transformer is selected for each group of ten houses, and the power factor is assumed to be 0.98 [38]. Thus, the July peak is 38.8 kV·A, which is equivalent to 155% loading. With no EV charging, F_{EQA} is calculated to be 1.37 in July in Los Angeles (LA) and 1.07 in July in Detroit. Winter F_{EQA} is well below one in both the cities, and thus, the annual F_{EQA} is less than one for both cities. Thus, a 25 kV·A transformer is reasonably sized for the residential load requirements. This article uses a transformer sized well for the residential load and not oversized for potential future EV charging needs, as this is the most critical concern in current-day distribution systems, as EV penetration rates begin to rise.

C. Solar and Temperature Data

This article investigates the benefits of solar-charged EVs using annual solar radiation and temperature data for two major U.S. cities: LA and Detroit. The solar radiation data are obtained from the 1998–2014 National Solar Radiation Database produced by the National Renewable Energy Laboratory using the Physical Solar Model (PSM) [39]. Figs. 4 and 5 show the annual solar radiation data for LA and Detroit, respectively, both for the case of full sun (as an upper bound) and for the average cloud case. Daily solar radiation profiles are averaged over a month for 21 years (1998–2018) to get a single profile to represent the average daily radiation profile of that month, as shown in Figs. 4 and 5. On sunny days, peak solar radiation in Detroit is only 7% less than peak solar radiation in LA. However, on days with average cloud, peak Detroit solar radiation is 19% less than that in LA. Monthly ambient temperatures of LA and Detroit, which are used in the transformer aging model, are obtained from [40]

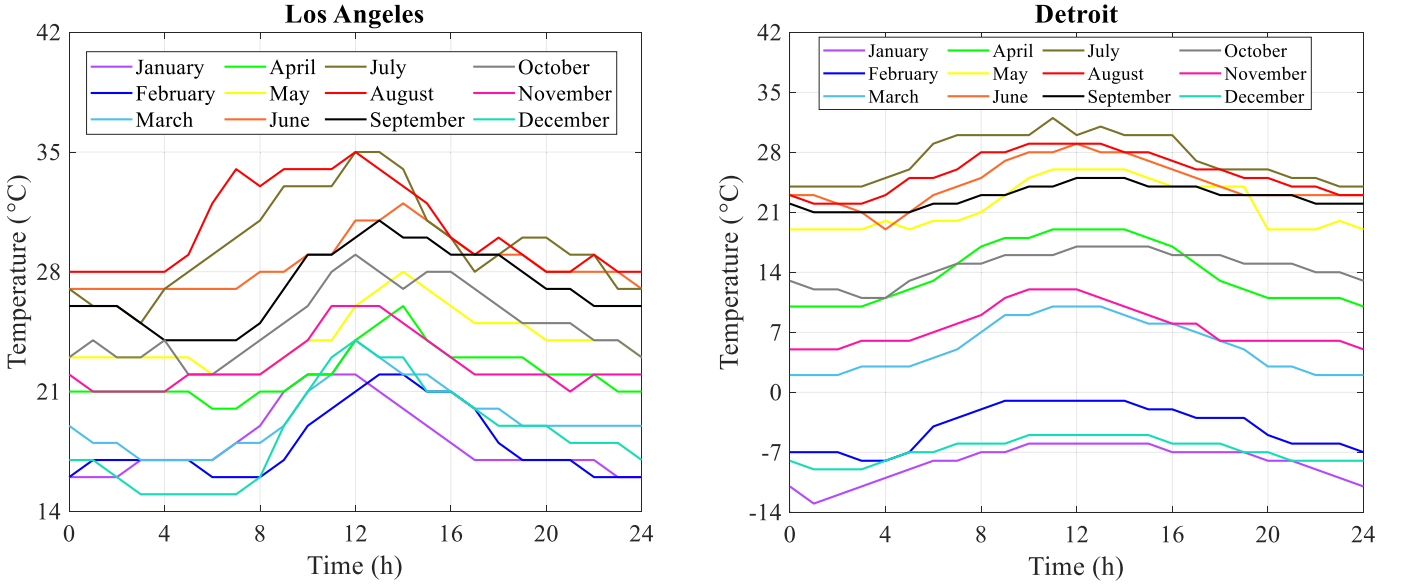


Fig. 6. Average daily ambient temperature profile for each month of the year in LA and Detroit.

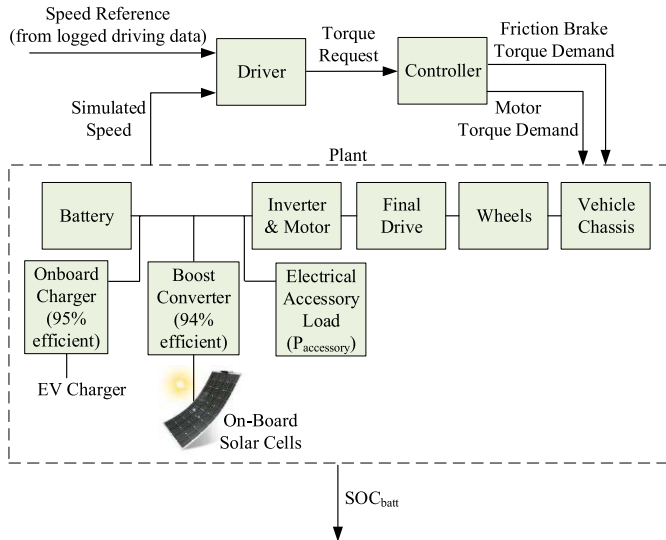


Fig. 7. Block diagram of SEV vehicle model.

and are shown in Fig. 6. The temperature profiles for each month represent the average daily temperature profile of the respective month.

III. SEV MODELING

A. Vehicle Modeling

A forward-looking vehicle model of the Chevrolet Bolt EV was created in MATLAB/Simulink, as described as follows. The nonsolar EV version was validated to be within 1.6% of EPA energy usage from dynamometer testing on city and highway drive cycles, as described in [41]. Fig. 7 shows a block diagram of the model with the proposed onboard solar assembly included. The onboard solar cells are further discussed in Section III-B, and the solar boost converter and associated power electronics are further discussed in Section III-D. The driver is modeled as a simple PI loop so that the driver torque

request is adjusted to follow the speed reference, which is input from the logged drive cycles. The controller block creates the motor torque command and the friction brake command subjected to the driver torque request and the motor speed–torque limits. In the plant model, the vehicle speed (in m/s) at the next simulation step, $v_{chas}(t+1)$, is calculated from the force out of the wheel block and the chassis aerodynamic losses, as shown in the following equation:

$$v_{chas}(t+1) = v_{chas}(t) + \left(\frac{1}{m} \int_t^{t+1} \left(F_{out_wheel} - \frac{1}{2} \rho_{air} A C_d v_{chas}^2(t) dt \right) \right) \quad (1)$$

where m is the vehicle mass in kg, ρ_{air} is the air density (1.23 kg/m^3), A is the vehicle frontal area in m^2 , and C_d is the coefficient of drag. The force out of the wheel block is calculated from the torque into the wheel block (τ_{in_wheel}), the friction braking torque ($\tau_{friction_brake}$), and the rolling resistance losses, as shown in the following equation:

$$F_{out_wheel} = \frac{\tau_{in_wheel} + \tau_{friction_brake}}{r_{wheel}} - (\mu_1 + \mu_2 v_{chas}) mg \quad (2)$$

where r_{wheel} is the wheel radius in meters, μ_1 and μ_2 are the rolling resistance coefficients, and g is the gravitational acceleration (9.81 m/s^2). The torque into the wheel block is equal to the motor output torque (τ_{motor}) multiplied by the final drive ratio (r_{fd}) and the final drive efficiency (η_{fd}), as shown in the following equation:

$$\tau_{in_wheel} = \tau_{motor} r_{fd} \eta_{fd} \quad (3)$$

The motor speed in rad/s is calculated using the following equation:

$$\omega_{motor} = \frac{v_{chas} r_{fd}}{r_{wheel}} \quad (4)$$

The motor and inverter are modeled as a lumped 2-D efficiency table, with the inputs of motor speed and motor

torque. The motor/inverter block calculates the DC input current required from the battery using the following equation:

$$I_{in_motor} = \frac{\omega_{motor} \tau_{motor} \times \eta_{motor}(\omega_{motor}, \tau_{motor})}{V_{batt}}. \quad (5)$$

The total battery current (I_{batt}) is the sum of the required motor current (I_{in_motor}) and the required electrical accessory current minus the current provided by the solar DC/DC boost converter (with efficiency $\eta_{solar_DC/DC} = 94\%$), as shown in (6). It is assumed that solar energy is available when the vehicle is driving or stationary

$$I_{batt} = I_{in_motor} + \frac{P_{accessory}}{V_{batt}} - \frac{P_{solar} \eta_{solar_DC/DC}}{V_{batt}}. \quad (6)$$

The battery model uses the battery current and an initial state-of-charge (SOC) value to determine the SOC and battery terminal voltage at the next simulation step. The SOC is the integral of the battery current divided by the total battery capacity (C_{batt}) as shown in (7). The battery open-circuit voltage (V_{batt_oc}) is determined from a lookup table (LUT) using the current SOC value. The terminal voltage (V_{batt}) is calculated from V_{batt_oc} and the internal battery resistance R_{batt} , as shown in (8). The Bolt EV has a 60-kWh battery, yet the usable energy is somewhat less. This article approximates the usable energy as 90% of the total energy, meaning that 54 kWh of energy is available between 5% and 95% SOC. Two grid-connected charging methods are considered: 1) charging is assumed to start at home after the last trip of the day, known as charge right away (CRA), and 2) charging is delayed until the overnight and early morning times using an available in-vehicle charging method called charge by departure (CBD). All charging is assumed to occur at a rate of 6.6 kW with an onboard charger efficiency of 95%

$$SOC(t+1) = SOC(t) + \frac{1}{C_{batt}} \int_t^{t+1} -I_{batt} dt \quad (7)$$

$$V_{batt} = V_{batt_oc} - I_{batt} R_{batt}. \quad (8)$$

Although vehicle heating and air conditioning (HVAC) loads will vary driver to driver due to personal preferences and weather conditions, it is important to create a model that realistically captures the differences in vehicle energy use across seasons and locations. The maximum continuous heating power is estimated at 4 kW, and the maximum continuous air conditioning power is estimated at 2 kW. In order to get the realistic values of vehicle energy, an LUT (Fig. 8) is added to the Bolt model to approximate HVAC loads based on the external ambient temperature data. An additional 300-W load is added to the HVAC loads for other accessories such as controllers, lights, and power steering.

B. Onboard Solar Cell Modeling

For the solar cell modeling and analysis of the proposed SEV concept, this section uses parameters from currently available technology to evaluate the plausibility and benefit of a fully integrated SEV, with the understanding that the work remains to be done to demonstrate its technical and economic viability. While fixed PV installations are typically installed tilted at a latitude in a southward facing direction,

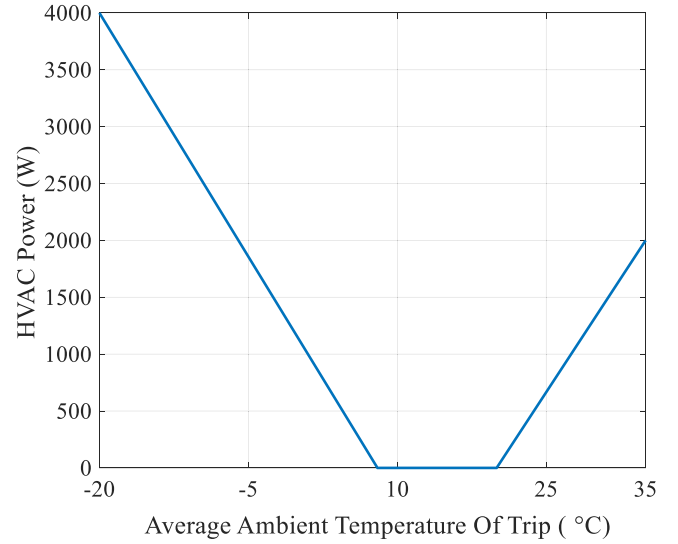


Fig. 8. Modeled relation between HVAC power use and trip average ambient temperature.

a mobile SEV creates new and unique challenges. For an SEV, all directions must be considered equally likely, leading to a strong preference for horizontal and near-horizontal surfaces in overall energy capture. However, since vehicles have body panel sections tilted at various angles from horizontal to vertical, the effect of panel tilt on energy capture is an important design consideration. At a high level, the proposed approach is to calculate the energy capture for a certain panel tilt, θ_t , and then convert this to an energy capture ratio relative to a horizontal (zero tilt) panel. In order to estimate the effect of panel tilt on the energy captured by an SEV panel, the calculation of the energy impinging on a tilted solar panel must be averaged over all panel rotation angles Φ because it is equally likely that a vehicle is facing any direction relative to the sun. This corresponds to a uniform distribution of vehicle orientations. The first step is to calculate the angle Ψ , between the panel normal and the solar direction, and find based on spherical trigonometry that

$$\cos \Psi = \sin \theta_t \sin \theta_z \sin \Phi + \cos \theta_t \cos \theta_z. \quad (9)$$

The function $\cos \Psi$ is the projection of the incident solar flux onto the tilted solar panel. Referring to Fig. 9, the solar direction is given by the solar zenith angle, θ_z , and the solar azimuth angle (not shown). The panel direction is given by its tilt with respect to the horizontal, θ_t , and its rotation angle, Φ , around the zenith. The solar zenith angle is the angle between the sun's direction and the zenith, a fictitious point infinitely far away, normal to the earth's surface at the position under consideration. With this definition, the zenith angle is 0° when the sun is directly overhead and 90° when the sun is at the horizon. The solar azimuth angle is the rotation angle around the zenith direction, relative to an arbitrary reference angle (such as north), of the vector from the observer to the sun. This angle is not required in the present calculations because of the angle averaging over Φ .

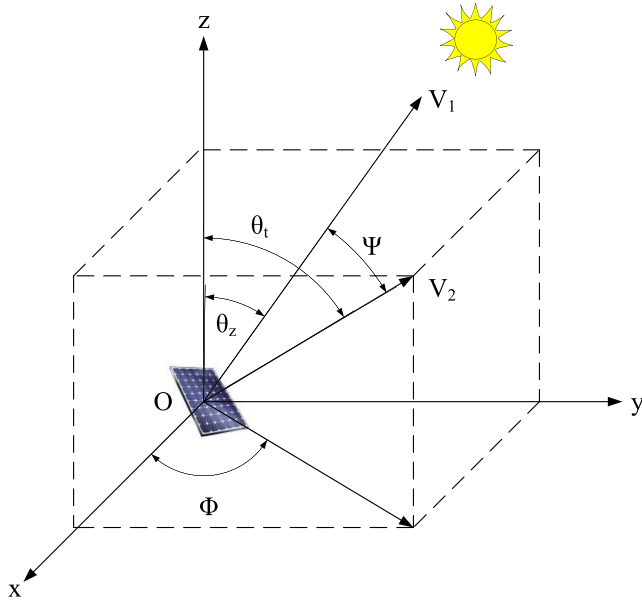


Fig. 9. Diagram of the spherical coordinate system used for calculating the projection cosine, $\cos \Psi$. The vector $\vec{V}_1$ points from the observer, O , toward the sun and is tilted by an angle θ_z from the zenith direction. The vector $\vec{V}_2$ is normal to the solar panel, which is tilted by an angle θ_t from the horizontal and rotated by an angle Φ around the zenith. The angle Ψ is the angle between $\vec{V}_1$ and $\vec{V}_2$.

The projection cosine, $\cos \Psi$, is then averaged over all angles Φ such that the sun is impinging on the front side of the solar panel to give a function $C(\theta_z, \theta_t)$. Next, for a given latitude, θ_L , the distribution function, $P(\theta_z, \theta_L)$, is calculated for all zenith angles throughout the year (for $0 < \theta_z < 90^\circ$) to weight the projection cosine average in arriving at an annual average. Since the air mass increases as the zenith angle increases, and with it the optical absorption increases, the optical transmission versus zenith angle is also used as a weighting function. Kasten's and Young's formulation [42] for the air mass dependence on zenith angle is used, with a function $AM(\theta_z)$, and Meinel's and Meinel's treatment [43] of the dependence of the optical transmission with a function $T(AM)$ is also used. Together, this gives an expression for the annual average solar energy impinging on a tilted solar panel, averaging over all panel presentation angles (because the vehicle is equally likely to face in any direction), as shown in the following equation:

$$E(\theta_t, \theta_L) = \frac{2}{\pi} \int_0^{\frac{\pi}{2}} C(\theta_z, \theta_t) \cdot P(\theta_z, \theta_L) \cdot T(AM(\theta_z)) d\theta_z. \quad (10)$$

For simplicity, the assumption is made that all solar illumination is direct, neglecting the more complex dependence of the small diffuse component on the zenith angle and also neglecting variations in aerosol and other atmospheric components. Finally, to facilitate the correction of modeled global horizontal irradiance, the energy capture, $E(\theta_t, \theta_L)$, is referenced to that for an untilted panel, $E(0, \theta_L)$, with the results as shown in Fig. 10. While this treatment does not include cloud cover, as a relative quantity, it is expected that this energy capture will provide a reasonable estimate of the dependence on panel tilt.

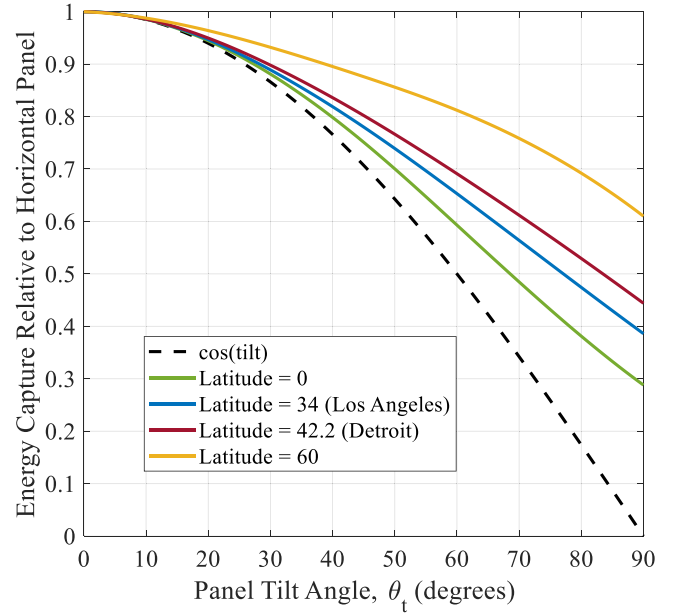


Fig. 10. Energy capture on an annual basis versus tilt of an SEV panel (relative to horizontal) and versus latitude, with latitude legend in degrees. The energy capture is referenced to the values for horizontal panels (i.e., 0° tilt).



Fig. 11. Reference Chevrolet Bolt as SEV.

Fig. 10 shows that for any latitude and panel tilt, the relative energy capture exceeds the $\cos \theta_t$ dependence shown for reference. At low tilt angles, the decreased power when the panel rotation is such that the panel is tilted further away from the sun is exactly compensated for by the increased power when the panel rotation is such that the panel is tilted toward the sun, leading to a slow (cosine) variation with the panel tilt angle. At higher tilt angles, the energy capture falls more slowly than a cosine function with a tilt angle because the sun is at high zenith angles that impinge on the tilted solar panel at many times of the day and year. While the absolute energy capture (not shown) decreases with latitude, the relative energy capture as shown in Fig. 10 increases with latitude, since the average zenith angle increases with latitude, making the tilted panels relatively more effective.

In this article, the Chevrolet Bolt is used as the reference vehicle, as shown in Fig. 11. In order to apply the results obtained in Fig. 10 to the vehicle model, equivalent horizontal

solar surface areas are calculated for each city studied and used in the main simulation. The physical dimensions of the Bolt show an approximately horizontal surface area of 2.33 m² (most of the roof) and an 18° slanted surface area of about 1.05 m² (including the hood and a slanted portion at the front of the roof). For the locations considered in this article, the relative energy capture for a tilt of 18° is 0.9566 in LA (downrating of 4.3%) and 0.9583 in Detroit (downrating of 4.2%). For the Bolt, an equivalent effective surface area is calculated as 3.334 m² in LA and 3.336 m² in Detroit, corresponding to total downrating factors of 1.35% and 1.30%, respectively.

This calculation also permits the evaluation of using vertical solar panels in an SEV context. Fig. 10 shows that the relative energy capture for vertical panels is 0.386 for LA and 0.444 for Detroit. Due to the significantly lower energy capture for vertical panels and their greater susceptibility to shading in normal traffic conditions, their inclusion would not be justifiable from a cost standpoint and thus have been omitted in the proposed SEV concept. Similarly, it is possible to consider the integration of a transparent PV module into the vehicle's front windshield, but the lost power in the visible part of the spectrum would greatly diminish the potential benefit. Finally, a comparison of the energy capture of a horizontal SEV panel to an optimally tilted fixed PV installation gives a ratio of 0.88 in LA and 0.81 in Detroit, indicating that SEVs would make a contribution comparable to rooftop installations on a per area basis.

The global PV market is dominated by flat PV modules comprised of monocrystalline or multicrystalline silicon solar cells which achieve high efficiency at low cost. Thin-film solar cell technologies are also implemented in flat modules for utility-scale deployment. For the SEV application, this article proposes that a thin-film technology with proven mechanical flexibility is the better choice to take the full advantage of the benefits of PV integration in an SEV. While other thin-film options may emerge in the future, this article suggests that CIGS cells are currently the best available choice due to their flexibility, proven ability to be integrated onto steel, and high efficiency [44]. With respect to other thin-film PV technologies currently available, CdTe is unlikely to be permitted [45] in new automotive applications due to the chemical toxicity of cadmium, whereas amorphous silicon has a substantially lower efficiency than CIGS. CIGS cells are manufactured in moderately high volumes, and flexible versions have been successfully deployed for roof applications. A record energy conversion efficiency of 23.35% has recently been demonstrated [19], [20] for a research-scale 1.043 cm² area, whereas 19.2% for a fully integrated module, 841 cm² in area [46]. Based on the record performance module, for illustrative purposes, typical module performance would be $V_{oc} = 0.7$ V, $J_{sc} = 38$ mA/cm², and $FF = 75\%$ under the standard AM1.5G testing spectrum at 25 °C, where J_{sc} is the short circuit current density, and V_{oc} is the open-circuit voltage. The fill factor FF is defined by $FF = (J_m V_m) / (J_{sc} V_{oc})$, where V_m and J_m are the voltage and current densities at the maximum power point, respectively, such that

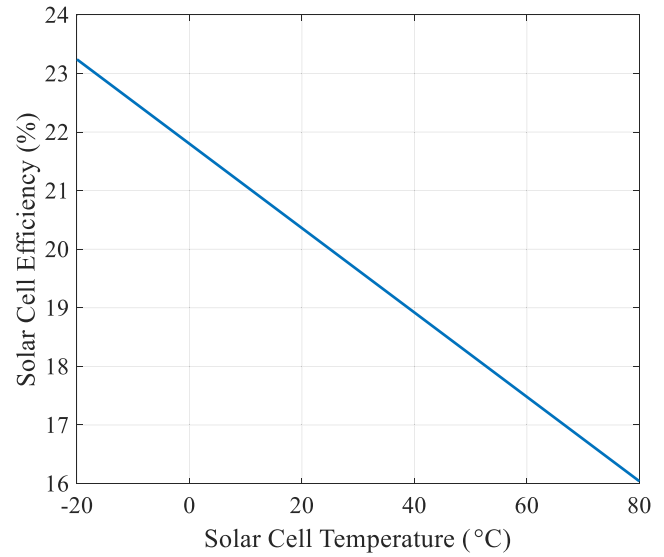


Fig. 12. Estimated CIGS cell efficiency as a function of cell temperature.

the efficiency, $\eta_{solar} = J_m V_m / P_0$, where P_0 is the incident solar flux.

PV efficiency decreases as cell temperature increases. CIGS cells have a relative efficiency downrating of 0.36% per °C, referenced to 25 °C [47]. In this article, 20% module efficiency (η_{solar}) at the nominal 25 °C is assumed as a realistic number attainable over the next 5 years. The resulting PV efficiency as a function of cell temperature is shown in Fig. 12. The cell temperature is calculated based on the ambient temperature (T_{Air}) and the solar insolation (S) in W/m², as shown in (11) [48]. In order to find an approximate peak solar power for sizing the electrical system, P_{solar} , S should be set high such as 1000 W/m², and T_{Air} should be set low, such as 10 °C. Using the effective horizontal solar area of 3.334 m² in LA, this results with a cell efficiency of 18.6%, with peak generated solar power, P_{solar} , equal to 620 W

$$T_{Cell} = T_{Air} + 0.035S. \quad (11)$$

As a fairly mature technology, the environmental impact of CIGS cell manufacture and deployment through its life cycle is well understood [49]. While the cells are comprised of gallium and indium, which are neither inexpensive nor earth abundant, their use in a thin-film format is believed to be feasible for SEV applications even at a global scale [50], particularly if CIGS is not widely deployed in the utility-scale market and if recycling is implemented. Selenium is also moderately rare, and since its compounds are toxic in high doses, it must be recycled and disposed of with appropriate caution. CIGS cells commonly include a very thin layer of CdS but can be made cadmium and lead free. Solar panels of any type have not yet been in production for long enough at scale to have an established recycling protocol in all countries [51]. Outside of Europe, the discarded PV panels are considered as “general waste,” whereas within Europe, panel disposal falls under the Waste Electrical and Electronic Equipment (WEEE) Directive [52]. Laws and regulations governing the end of

life recycling of automobiles are expected to be sufficient to ensure that CIGS cells are properly recycled or disposed, such as the European Union's End-of-Life Vehicles (ELV) Directive [53]. Recycling of the rare elements in CIGS cells has been demonstrated, and due to the high cost of gallium and indium, it is considered economically very advantageous [54]. Via accelerated testing protocols, CIGS cells have been demonstrated to have usable lifetimes in the field, with typical power degradation of less than 10% over 10 years and less than 20% over 25 years, a standard warranty requirement for utility-scale PV [55]. While high-temperature processing is used in the manufacture of CIGS cells, as a thin-film technology, the energy input is low, leading to an energy payback time of less than 1 year, depending on manufacturing process and location of deployment [56].

In the manufacture of a CIGS module, the device layers are deposited in a batch fabrication process, sectioned into multiple electrically isolated cells which are then interconnected into a series string via a set of patterning processes [57]. This architecture is generally referred to as monolithic cell interconnection and requires an insulating layer or substrate in order to enable electrical isolation of the cells. Without this sectioning, a single 1 m² CIGS cell would have $V_{oc} = 0.70$ V and $I_{sc} = 380$ A. By sectioning into 100 cells, with a series interconnection, this becomes a much more manageable $V_{oc} = 70$ V and $I_{sc} = 3.8$ A, which is typical for rooftop and utility-scale applications. Such series-connected panels are then combined in parallel to provide more overall power. There are several potential benefits to a greater level of sectioning for the SEV application under consideration. First, the voltage can be increased to be closer to the EV battery voltage (250–450 V), meaning that less voltage boost is needed in the DC/DC converter, and thus, higher DC/DC converter efficiency can be obtained. Second, by achieving a higher voltage and lower current, the losses associated with series resistances are reduced. This also permits the use of thinner wires, reducing cost and weight. A real advantage of thin-film PV is that there is a tremendous flexibility in the level of sectioning and even the opportunity to adapt the cell size and interconnection pattern to take the full advantage of curved surfaces on automotive sections. This is in contrast to silicon PV technology, where individual cells are batch fabricated in a fixed size format, such as 15×15 cm². A novel approach to boost voltage via series-connected PV strings is by stacking cells vertically with cell thicknesses adjusted to maintain a current-matched condition. This concept has been demonstrated in monolithically grown GaAs photoreceivers with stacks of over 20 lattice-matched cells [58], employed for telecommunications applications. For SEV applications, some optimization in cell size and interconnection pattern will be required to achieve the best overall energy yield. The flexibility provided by thin-film sectioning and/or cell stacking provides additional degrees of freedom to do so.

CIGS cells are most typically grown on a rigid glass substrate that also serves as a mechanical support for the PV module. However, commercial CIGS cells are also made on flexible substrates, using roll-to-roll manufacturing [57]. In order to implement monolithic cell interconnection as

described above, the flexible substrate must either be insulating or a flexible metal foil with an intervening insulating layer. In the latter case, the insulating layer can also serve as a source of sodium dopants and/or as a diffusion barrier from impurities in the metal substrate. High-quality CIGS cells have been grown on stainless steel using SiO₂, Al₂O₃, or Si₃N₄ insulating layers [57]. The record efficiency to date for CIGS on stainless steel is 20.56% for a research-scale 0.86 cm² area [59]. More recently, high-quality CIGS cells have also been demonstrated on mild steel substrates [60], suggesting that CIGS cells could be grown directly on automotive steel parts with suitably optimized layers for insulation, doping, corrosion protection, and inhibiting impurity diffusion. CIGS cell growth is typically followed by a polymer encapsulant such as EVA, providing long-term hermeticity. In an SEV context, the encapsulant needs to be replaced with an automotive paint that is hermetic, provides mechanical protection, and is optically transparent over the relevant part of the spectrum (~350–1100 nm for CIGS), which includes the entire visible spectrum. This constraint would likely limit the esthetic design flexibility and may limit consumer acceptance; however, creative marketing, such as what Tesla has done for EVs, in general, may minimize this issue. For a CIGS cell fully integrated onto the automotive steel and encapsulated as described, the incremental mass from the CIGS layers would be only ~40 g/m², for a total mass of 133 g. The overall mass increase is likely to be dominated by wiring, bypass diodes, and DC/DC conversion components, which are estimated to be less than 4 kg for a 700-W system.

This analysis assumes low-cost 20% efficient (nominal) CIGS thin-film PV cells because this approach is currently feasible for the proposed SEV application. However, the resulting benefits will increase if more efficient cells are used, such as GaAs thin-film PV cells with 29% efficiency. The manufacturing cost of GaAs thin-film cells is currently very high, but the cost reduction is an active area of III–V semiconductor PV research [61]. The high cost is due to contributions from an expensive GaAs growth substrate and the high cost of ownership of the metal–organic chemical vapor deposition (MOCVD) reactors used to grow the cells, due to the relatively slow growth process [62]. The substrate costs can be substantially mitigated by epitaxial lift-off from the parent wafer and subsequent wafer reuse. The MOCVD-related costs can be reduced by the development of higher growth rate processes [61], the use of larger wafers and larger batch sizes, and improved precursor utilization. Thus, higher efficiency GaAs thin-film cells may become viable options for SEVs in the future. Alta Devices, which holds the GaAs efficiency record [63], is currently developing GaAs cells for a number of high-end transportation applications.

C. Modeling of Partial Shading

The range of potential partial shading scenarios is wide and will depend on each driver's unique environment (urban, country, nearby buildings, and/or trees). Thus, this article will study one representative case that can offer insight into the effects of typical shading scenarios. In this case, a vehicle is parked 5 ft away from a 15-ft-tall structure, where the vehicle

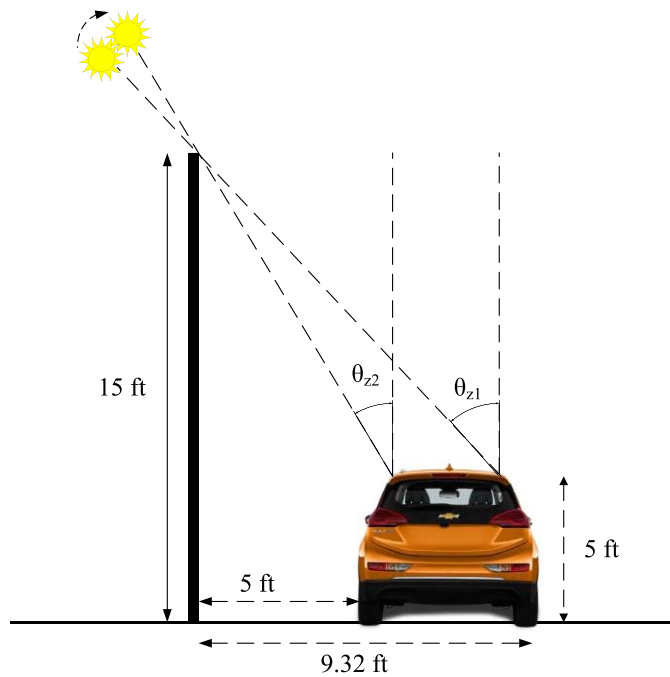


Fig. 13. Example case for partial shading modeling (structure on the left side of vehicle).

and structure direction are set to the worst case, where the sun rises directly behind the structure. To represent changing vehicle positions over a day, the case is modeled such that, in each hour of the day, the structure is on the left side of the vehicle for 20 min, the structure is on the right side of the vehicle for 20 min, and the structure is absent for 20 min. This pattern is repeated through the day as the sun changes position.

Fig. 13 shows the case with the structure on the left side of the vehicle, where the average height of the vehicle's solar panels is estimated at 5 ft. The sun will start shining on the far side of the vehicle when the zenith angle, θ_{z1} , decreases to 42.98° . As the sun moves, the radiation on the vehicle solar panels will increase—the solar panels will be in full sun when the zenith angle, θ_{z2} , decreases to 26.57° . This analysis leads to two important findings: 1) since the peak solar radiation occurs around midday, when the zenith angle is the lowest, partial shading will have little or no impact on the solar energy captured during this peak solar energy time (i.e., when the sun is high in the sky, more directly above the vehicle) and 2) since the zenith angle does not reach 26.57° in the winter and some spring and fall months of the year (as shown in Fig. 14), there will be zero solar energy captured in these months when the structure is between the sun and the vehicle.

When the partial shading hourly pattern is repeated as described over a day, the solar radiation pattern becomes choppy and variable, as shown in Fig. 15. The monthly solar energy generated decreases the most in the winter months due to the high zenith angles—by about 33% in January in both LA and Detroit. However, in July, the solar energy reductions are only 13.8% and 17.3% in LA and Detroit, respectively, due to the lower zenith angles of the sun. The solar radiation curves

shown in Fig. 15 are based on average cloud conditions and will be used for the partial shading simulations in this article.

D. SEV Electrical Architecture and Challenges

Although this article focuses on the system-level benefits of SEVs to investigate if further detailed engineering work is warranted, it is also important to understand the technical challenges of designing a low-cost and high-efficiency SEV electrical architecture. The optimal design of the electrical architecture for an SEV is a multifaceted problem because it depends on the PV cell size and connection scheme, the PV cell voltages, and the power electronic converters, where optimal converter choices can change depending on the first two variables. This section will discuss the design issues on the path toward a low-cost, low-mass, and high-efficiency SEV electrical architecture.

With regard to PV cell size, the proposed thin-film CIGS material can be used to effectively make cells of various sizes. Larger cell sizes mean that fewer interconnections are needed, but lower voltage is produced from a string of cells connected in series. A related challenge is to decide how many cells should be connected in series and parallel. More cells in series connection boost the resulting voltage, but if any shading occurs in the series string, the solar power will be reduced for the entire string. Since much of the harvested solar energy may be obtained while the vehicle is parked, an optimal design should allow for maximum solar power production even when parts of the vehicle body are in the shade. However, if fewer cells are connected in series, the low resulting voltage will require a high boost to the traction battery voltage, which generally leads to poor efficiency. The previous research on MPPT for partially shaded PV cells indicates that the most common methods for dealing with partially shaded cells are to use bypass diodes (low cost) or to add DC/DC converters to each cell (higher efficiency and higher cost) [64]–[66]. Also related to MPPT is the issue of the slightly curved nature of some vehicle surfaces, such as the hood. Since PV panels are generally considered as flat surfaces, further research is required to determine how to extract the maximum solar power from PV cells distributed over a curved surface. However, the problem is similar to that of partial shading in that each cell along the curve will have a different maximum powerpoint, so similar techniques may be able to be applied to this problem. In this article, it is assumed that MPPT is used in the system, meaning that the full available solar energy can be captured.

The main power electronic challenge is to boost the PV voltage to the high traction battery voltage, which commonly varies between 250 and 450 V in EVs. Although a 12-V battery is used in EVs for low-voltage accessories, the energy capacity of these often lead-acid batteries is not sufficient to store solar energy gathered over a long period of time. Many architectures can be considered, and Fig. 16 shows the two potential strategies: 1) keep the PV voltage low (by using a low number of cells in series) and reuse the bidirectional high voltage to 12-V DC/DC converter already present in most EVs or 2) create high PV voltages using

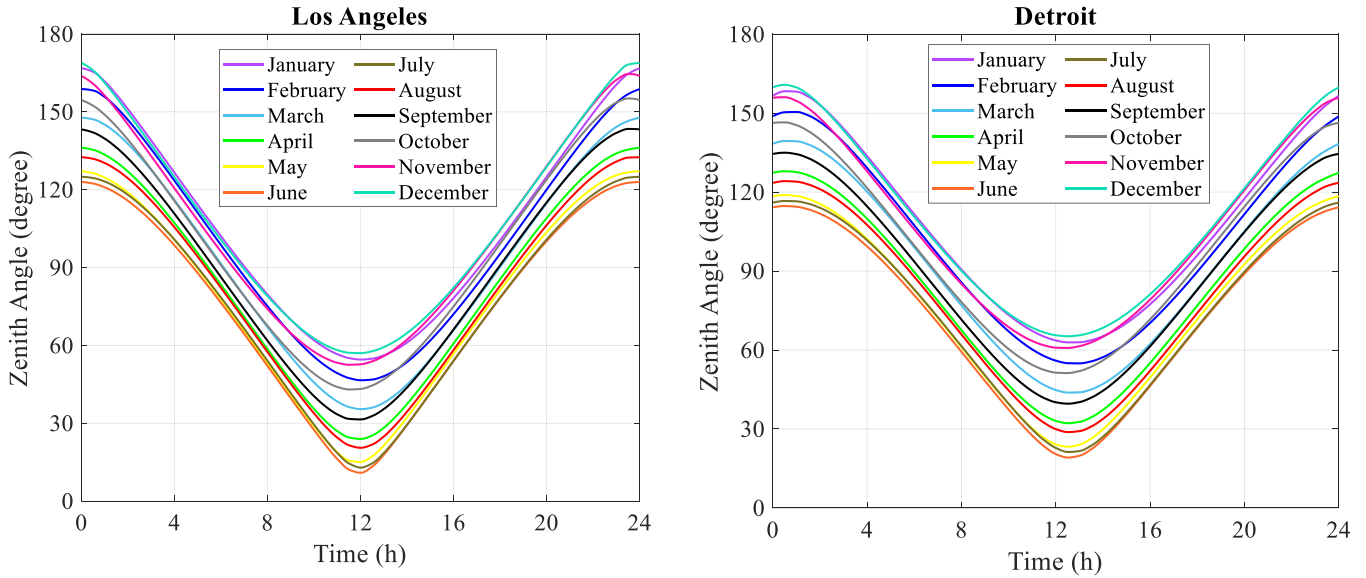


Fig. 14. Monthly average zenith angle profile for LA and Detroit.

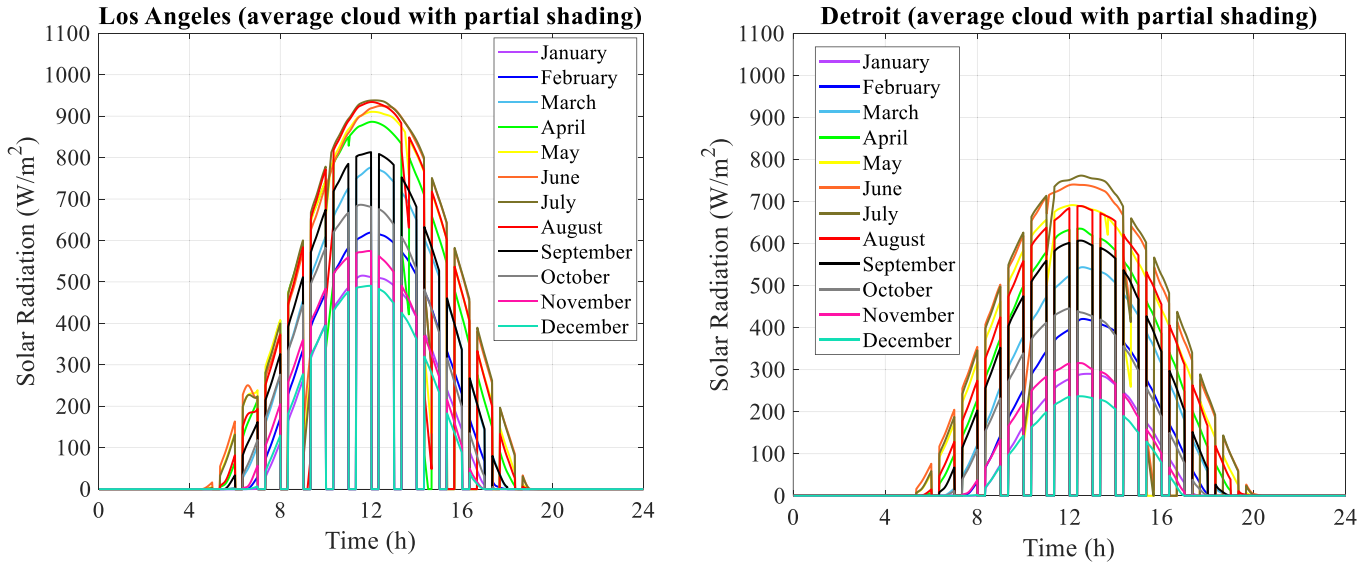


Fig. 15. Monthly partial shading solar radiation models for LA and Detroit.

vertically stacked cells and/or more cells connected in series and then boost the voltage to the traction battery voltage using dedicated DC/DC converters. The advantage of Option 1 is that cost is reduced by reusing the existing DC/DC converter for the main voltage boost to the traction battery voltage. The disadvantage of Option 1 is that lower efficiency is expected due to: 1) higher currents in each MPPT converter due to the lower input voltage; 2) the existing bidirectional DC/DC converter is usually sized between 2 and 4 kW and thus may have poor efficiency at the 200–620 W level of solar power in the proposed SEV; and 3) solar power must be processed by two DC/DC converters before reaching the traction battery. The advantages of Option 2 are: 1) higher voltages are used, generally leading to higher efficiency, and 2) the dedicated DC/DC converters will be higher efficiency due to less voltage

boost required and the possibility to optimize the design solely based on the lower solar power levels. Still, the high step-up MPPT boost converters in Option 2 will add additional cost compared to Option 1. For a system-level analysis, it is not critical to design the details of the SEV electrical architecture, yet a reasonable approximation of losses in the architecture must be used so that harvested solar energy is correctly estimated. In this article, Option 2 is assumed, and the total efficiency of the power electronic converters in the path from the PV cells to the traction battery is estimated at 94% based on similar high step-up boost converters [67], [68].

E. Transformer Aging Model

The main concern in overloading distribution transformers with household and EV charging loads is accelerated

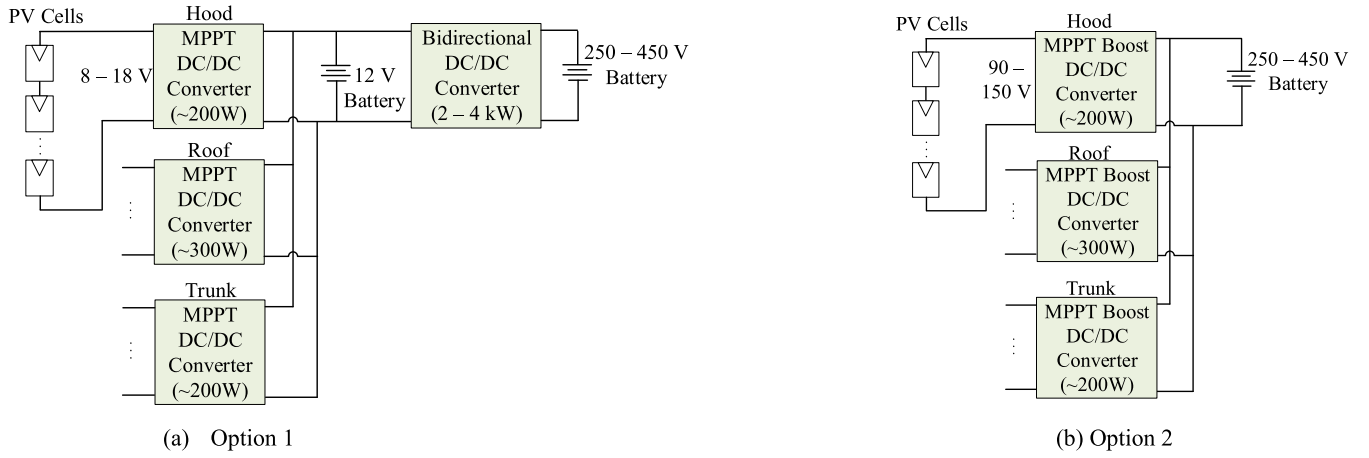


Fig. 16. Two potential electrical architecture options for the proposed SEV. (a) Option 1. (b) Option 2.

transformer aging. Thus, the equivalent aging factor, F_{EQA} , is a useful metric for comparing the EV charging effect on transformers. Transformer aging is mostly due to the degradation of the insulation; this depends on temperature, moisture, and oxygen content. In modern oil-cooled transformers, the effect of moisture and oxygen content on insulation deterioration can be neglected. Thus, transformer accelerated aging is mainly related to the hottest-spot insulation temperature [37]. This article uses the overhead distribution transformer aging model from IEEE C57.91-2011 [37], as summarized as follows. (Full details are given in [37].)

The total household load plus the simulated EV charging load are converted to an equivalent load, L_{equiv} , using the following equation:

$$L_{\text{equiv}} = \sqrt{\frac{L_1^2 \Delta t_1 + L_2^2 \Delta t_2 + \dots + L_n^2 \Delta t_n}{\Delta t_1 + \Delta t_2 + \dots + \Delta t_n}} \quad (12)$$

where $L_1, L_2, \dots, L_n$ are various kilovolt per ampere load values including the baseline household load and the EV charging load that has been simulated for that particular driver on each day of the logged week and for each month, where HVAC loads vary by the month. These $L_1, L_2, \dots, L_n$ loads are specified in 1-min intervals $\Delta t_1, \Delta t_2, \dots, \Delta t_n$, respectively, so that high-resolution charging patterns are used in the transformer aging analysis.

The hottest-spot temperature, θ_H , is the summation of three temperature components in degree Celsius [37] as shown in the following equation: ambient temperature, θ_A , delta between the top-oil temperature and the ambient, $\Delta\theta_{TO}$, and delta between the hottest-spot temperature and the top-oil temperature, $\Delta\theta_H$:

$$\theta_H = \theta_A + \Delta\theta_{TO} + \Delta\theta_H \quad (13)$$

where θ_H is directly related to the insulation life of the transformer. $\Delta\theta_{TO}$ and $\Delta\theta_H$ are dependent on L_{equiv} and other variables as described in [37]. F_{AA} is defined as the aging acceleration factor and is an exponential function of the hottest-spot temperature θ_H as shown in the following

equation:

$$F_{\text{AA}} = e^{\left(\frac{B}{383} - \frac{B}{\theta_H + 273}\right)}. \quad (14)$$

In (14), B is the cellulose aging rate constant, with a value of 15 000 based on the transformer insulation life curve in [37]. The transformer specification data were obtained from [69].

The equivalent aging factor, F_{EQA} , is a useful parameter because it represents how fast a transformer ages based on its thermal profiles compared to the normal condition. For example, $F_{\text{EQA}} = 1$ means that the transformer has aged 1 week over the test week period, and $F_{\text{EQA}} = 2$ means that the transformer has aged at double the normal rate or 2 weeks over the test week period. F_{EQA} is described in the following equation, where Δt_n is each time interval, N is the total number of time intervals, and $F_{\text{AA}n}$ is the aging acceleration factor during the time interval, Δt_n :

$$F_{\text{EQA}} = \frac{\sum_{n=1}^N F_{\text{AA}n} \Delta t_n}{\sum_{n=1}^N \Delta t_n}. \quad (15)$$

F. Simulation Process

The first step in the simulation process is to consolidate individual logged trips into driving days, where the end of the last trip before 12 A.M. constitutes the end of a driving day, with the next driving day starting on the next trip after 12 A.M. For each trip, the logged data give the actual date and start time of the trip—using this data, the correct “day of the week” is assigned to each driving day, as, for example, the driving patterns on a Wednesday will differ significantly from those on a Saturday. Thus, the logged driving data give a very accurate account of an actual week of driving for each of the 150 participants. For each driver, the Bolt EV model is simulated on the speed profiles of each trip of the logged days, and the simulated ending battery SOC of one trip is fed back into the vehicle model as the starting SOC for the next trip that day.

To create an annual analysis, this article assumes that the driving pattern of an individual will remain static across all

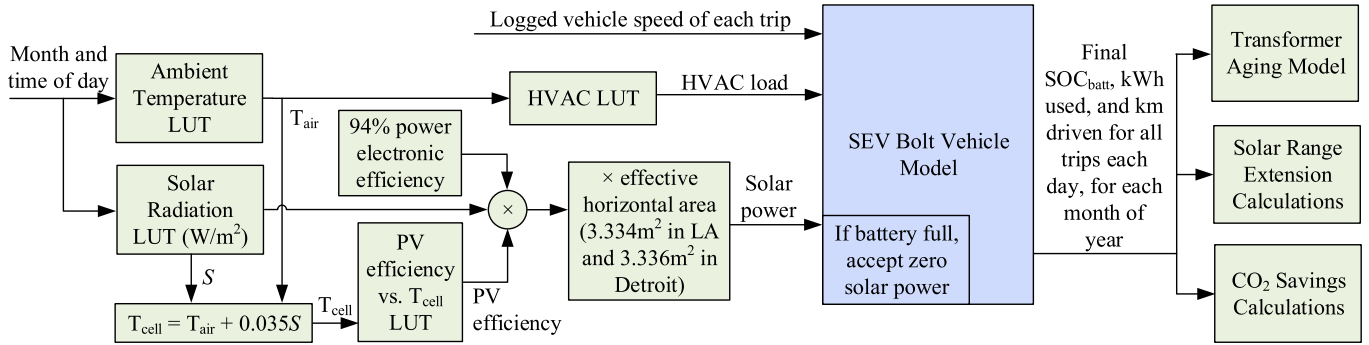


Fig. 17. Summary of simulation process. This process is looped: for 150 drivers, for each day of the week, for each month of the day, for full sun, average cloud, and average cloud with partial shading conditions, and in two cities. This is also then rerun with solar radiation = 0 to represent the nonsolar EV case.

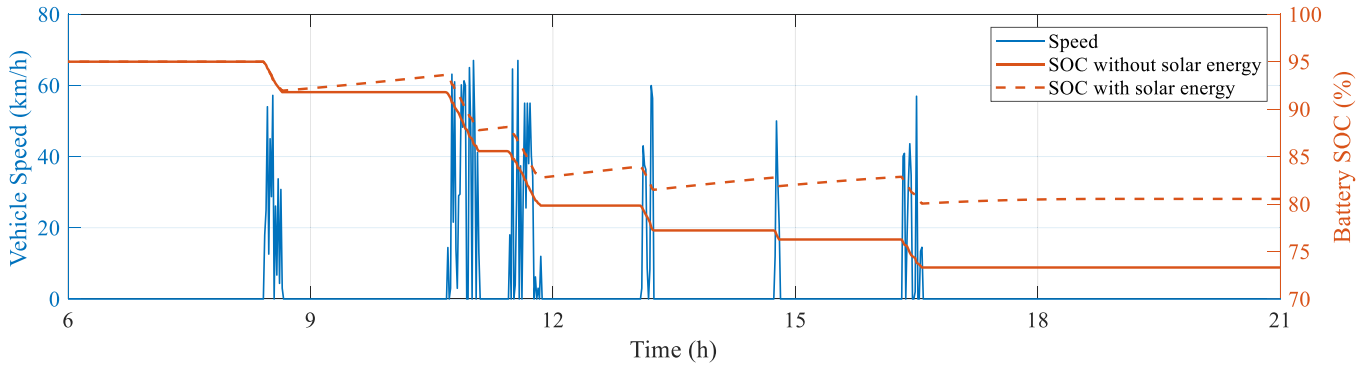


Fig. 18. Vehicle speed and battery SOC for driver #1 over one example day, both for an EV without on-board solar, and the proposed SEV.

weeks of the year. The simulation process is shown in Fig. 17, where “LUT” represents a lookup table. For each of the 12 months of the year, the driving week for each participant is simulated under ambient temperature and solar radiation conditions (full sun, average cloud, and average cloud with partial shading) for LA and Detroit. The result is 12 unique monthly values for vehicle energy used and grid charging energy needed for each driver in each city, which are used in the transformer aging model to obtain an annual transforming aging estimate. These simulation results are also used in the solar range extension and CO₂ savings calculations, as part of the driver and environmental analysis. For all simulations, a case is also run with solar radiation set to 0 to represent the nonsolar EV case.

To illustrate the simulation result details, Fig. 18 shows the driving pattern and resulting battery SOC of Driver #1 on one logged day, both with and without using onboard solar energy. The results show that the battery pack cannot start storing solar energy until the vehicle starts driving for the day (with the assumption that full charge is obtained from the grid overnight) and that significant solar energy can be stored during the parked times of the day thereafter. On a sunny July day, if both EVs start at 95% SOC, the SEV SOC ends at 80.54%, and the nonsolar EV SOC ends at 73.31%. Thus, at the end of the day, the SEV needs 8.68 kWh of charge from the grid and the nonsolar EV needs 13.01 kWh and a reduction of 4.33 kWh or 33.3%. Solar charging stops in the late afternoon due to sunset.

IV. SIMULATION RESULTS AND DISCUSSION

A. Energy Use

All grid, driver, and environmental benefits ultimately stem from the reduced net vehicle energy consumption of SEVs compared to nonsolar EVs. For the proposed large-scale integrated SEV concept in average cloud conditions, the annual vehicle energy consumption is reduced by 21.5% in LA and 17.5% in Detroit. For comparison purposes, the sun-rich location of Solar Village, Saudi Arabia (latitude 24.91° and relative energy capture 0.9555), is also analyzed as it has excellent solar radiation data available [70]. A combination of high incident solar energy and high ambient temperatures, as shown in Fig. 19 [40], [70], results in high vehicle HVAC energy use during most of the months and an annual vehicle energy consumption reduction of 22.7% in average cloud conditions compared to a nonsolar EV in Saudi Arabia. Furthermore, the analysis was also performed for LA using a small vehicle rooftop solar panel such as that on the Toyota Prius Prime [22]. With a peak claimed output of 180 W [22], the panel size is estimated at 1 m² assuming the same efficiency as the cells modeled in this article. For comparison purposes, this vehicle rooftop solar panel would result in 7.6% reduction in annual vehicle energy consumption in LA under average cloud conditions, compared to a nonsolar EV.

B. Grid Benefits

The main grid benefit of utilizing SEVs instead of nonsolar EVs is the reduction in load during peak times of the day—this

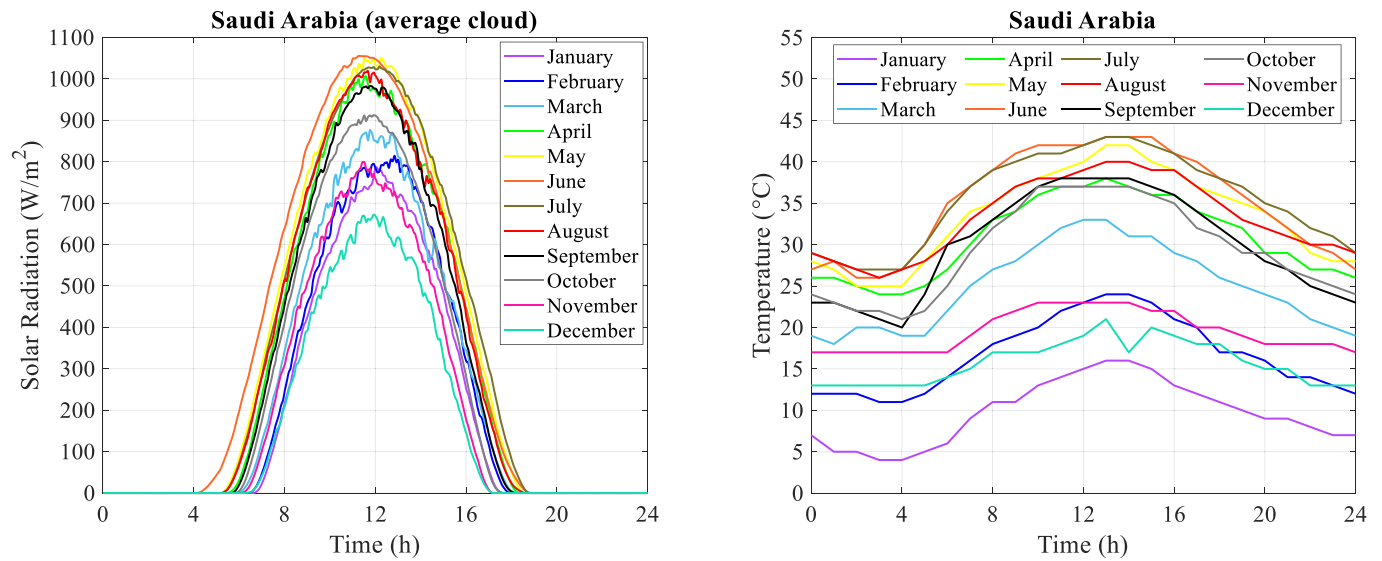


Fig. 19. Average daily radiation and ambient temperature profile for each month of the year in Saudi Arabia.

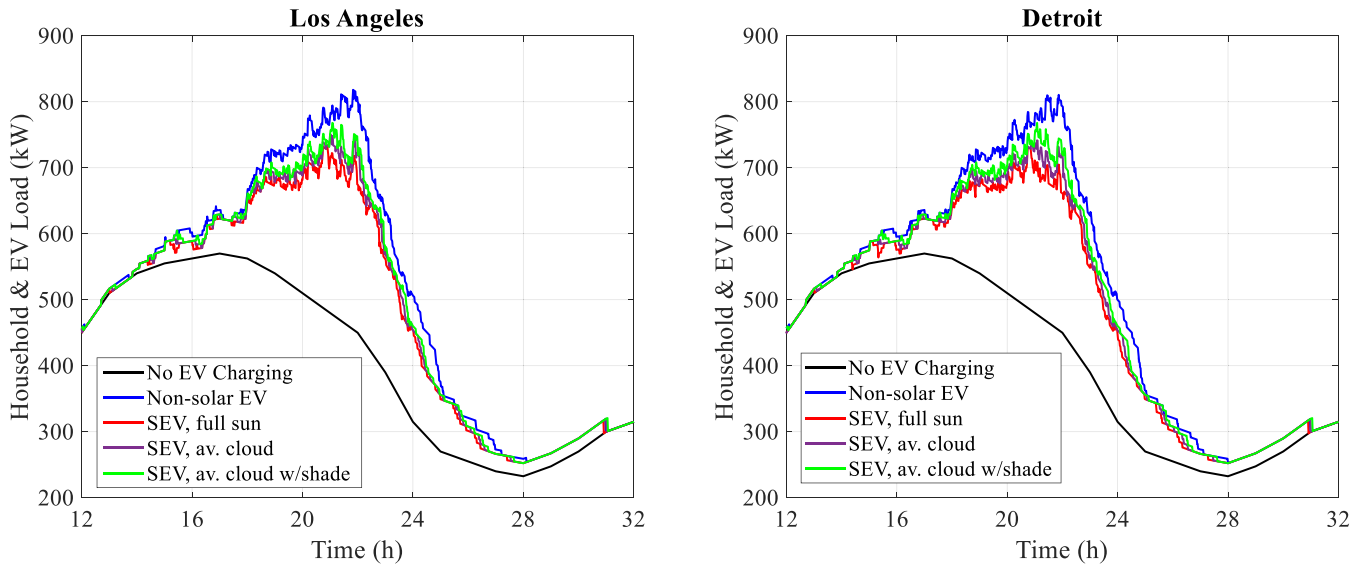


Fig. 20. Household and EV charging load for 150 EVs and 150 houses for nonsolar EV, SEV on a day with full sun, average cloud conditions, and average cloud conditions with partial shading (CRA charging).

will lead to lower the loads on distribution transformers (and thus, lower aging), lower loads on transmission lines (and thus, lower losses and lower potential for overload), and lower peak energy generation. This article focuses on modeling the reduction in the aging of distribution transformers with the use of SEVs as this is a pressing concern as EV adoption rises in neighborhoods with older distribution grids. EV charging has been simulated based on the simulated energy use of each logged driver and the unique arrival time home of each logged driver. The total load (residential + EV charging load) for 150 EV drivers (100% EV penetration) using CRA charging is shown in Fig. 20 for the logged Wednesday in July. The temperature profiles in this hot month are similar for both the cities, and therefore, the vehicle energy consumption due to HVAC is similar for both the cities. The total charging energy

consumed by 150 EVs at the plug in LA and Detroit on a typical Wednesday in July is 2.20 and 2.14 MWh, respectively. When these EVs are replaced by SEVs, the charging energy consumption is reduced to 1.61, 1.63, and 1.68 MWh for full sun, average cloud, and average cloud with partial shading cases in LA and 1.53, 1.59, and 1.66 MWh, for the same cases in Detroit, respectively. The peak power reduces from 817 to 735 kW in LA and from 810 to 732 kW in Detroit for the SEV full sun case. The main effect of the SEV is that although charging start times remain the same compared to the case of nonsolar EVs, the charging end times are earlier (though unique for each driver) due to the higher ending SOC of the vehicle at the end of the driving day for SEVs. As shown in Fig. 20, this reduced charging duration significantly lowers the total load in the evening.

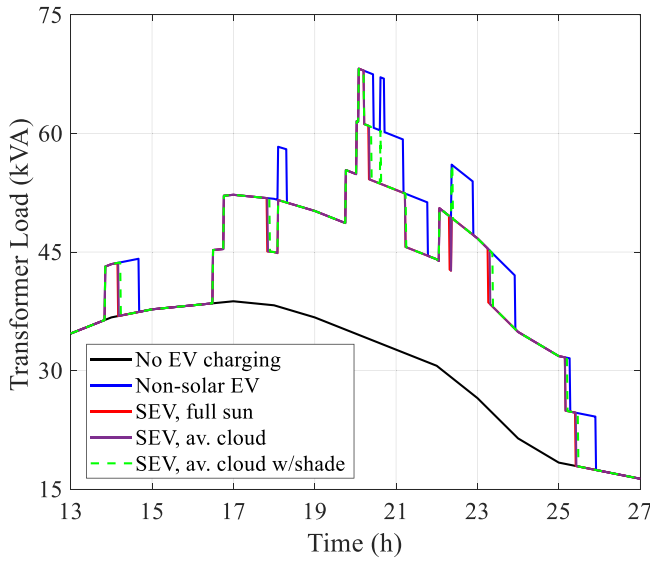


Fig. 21. Household and EV charging load for one transformer for nonsolar EV, SEV on a sunny day, SEV on a day with average clouds, and SEV on a day with average clouds and partial shading (CRA charging).

Fifteen distribution transformers are not evenly loaded as every EV has its own charging requirement. Fig. 21 shows the detailed charging profile of one transformer feeding ten homes simulated with ten SEVs and ten nonsolar EVs using CRA charging on a Wednesday for LA in July. Since the transformer aging equations for F_{EQA} are nonlinear, a moderate difference in the transformer charging energy makes a large difference in F_{EQA} . For example, in LA, the energy required from the grid to charge the ten nonsolar EVs on the simulated Wednesday is 168.5 kVAh, and the energy needed to charge the SEVs on the same day is 137.5 kVAh (full sun), 138.5 kVAh (average cloud), and 141.2 kVAh (average cloud with partial shading). Thus, the use of the SEVs has reduced grid energy charging needs by 18.4%, 17.8%, and 16.2%, respectively. However, when considering the transformer F_{EQA} for the same day, the nonsolar EV transformer has $F_{EQA} = 35$, and the SEV transformer has $F_{EQA} = 19.3$ (full sun), $F_{EQA} = 19.4$ (average cloud), and $F_{EQA} = 19.6$ (average cloud with partial shading) which is about a 45% reduction in transformer aging using SEVs for this day. The peak transformer power for the case of SEVs is equal or 6.6 kW less than for the case of the nonsolar EVs—thus, transformer aging is not solely dependent on peak power but has a more complex relationship with the whole charging profile and associated heating. A moderate reduction in charging energy during peak load times of the day results in a significant reduction in transformer aging.

Depending on the variable temperatures throughout the year as shown in Fig. 6, the vehicle HVAC energy consumption varies from month to month, and the EV charging energy thus varies accordingly. Furthermore, the grid energy generated at the generator is about 12% higher than the distribution demand at the point of load due to transmission and distribution losses [7]. Fig. 22 shows the generated grid energy required to charge 150 EVs per month for the cases considered in LA and Detroit. It can be seen that the required grid charging

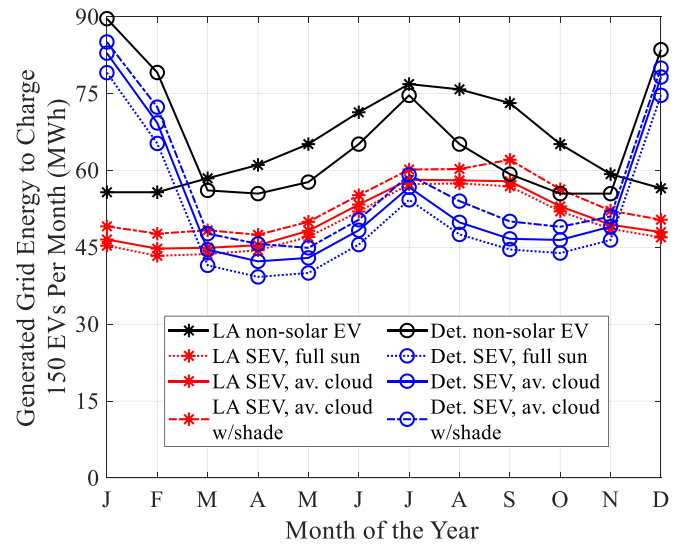


Fig. 22. Generated grid energy required to charge 150 EVs in each month of the year in LA and Detroit (CRA charging).

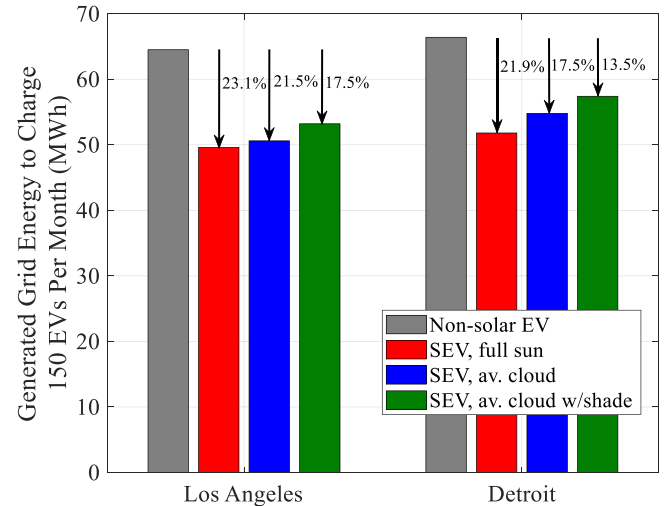


Fig. 23. Annual average monthly generated grid energy required to charge 150 EVs (CRA charging).

energy is the lowest in the winter months and highest in the summer months in LA. The reason is that the winters in LA are mild and generally do not require cabin heating, whereas the summers are hot and will often require the use of cabin air conditioning. Conversely, in Detroit, the winter months have the highest EV charging loads due to the high energy consumption required to heat the cabin in cold ambient temperature. In spring and fall, vehicle energy consumption decreases due to milder temperatures not requiring any cabin HVAC, and then cabin air conditioning increases the energy use again in the summer but to a lower value than the heating required in the winter. SEVs can reduce the energy generation demand in a sunny July month in LA by 25% and Detroit by 27%, which is an important time for reducing energy use due to the otherwise peaking grid load. The annual average monthly generated grid energy required to charge 150 EVs is shown in Fig. 23. In the presence of moderate partial shading,

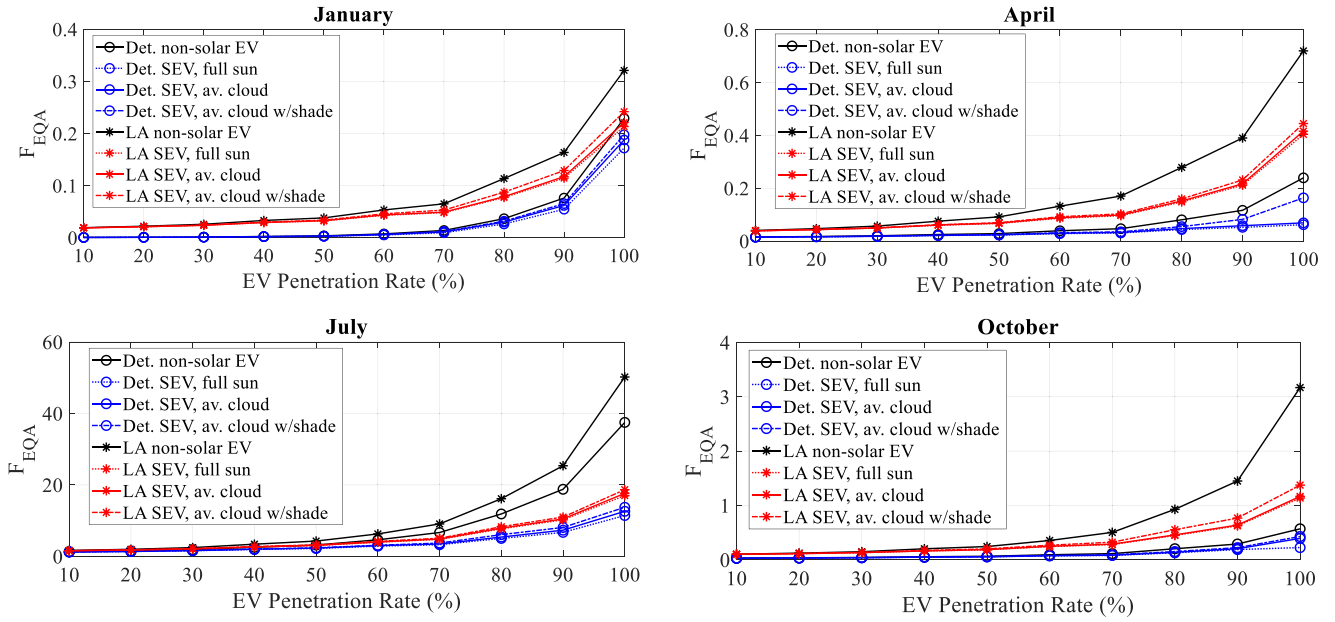


Fig. 24. Average monthly equivalent aging factors for nonsolar EVs and SEVs in Detroit and LA (CRA charging).

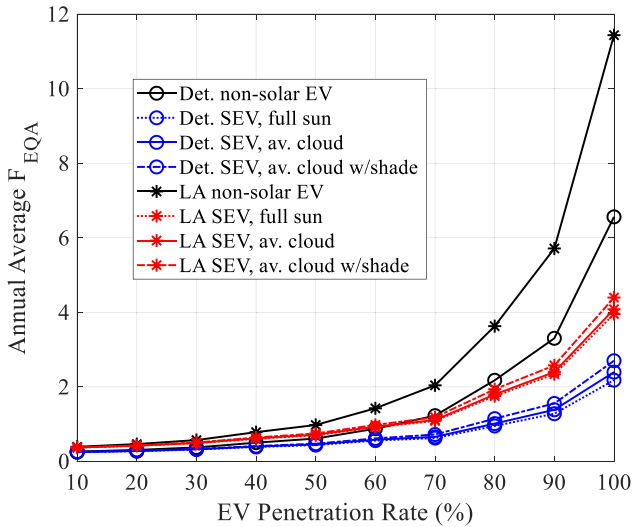


Fig. 25. Average annual equivalent aging factors for nonsolar EVs and SEVs in Detroit and LA (CRA charging).

the average reduction in grid energy generation is 17.5% in LA and 13.5% in Detroit.

Based on the charging profiles for each day of the week, the equivalent aging factors, F_{EQA} , are calculated for each of the 15 transformers in each city and each month of the year. The average monthly aging factors for all 15 transformers for 4 months for each EV penetration rate are shown in Fig. 24. Fig. 24 shows that for nonsolar EVs, the equivalent transformer aging factor is less than 1 for 100% EV penetration in the colder month of January, for both LA and Detroit. Thus, the cooler ambient temperatures are a larger factor in transformer aging than the high charging energy needed in the coldest months due to vehicle HVAC use. In the warmer months, SEVs offer a significant improvement in transformer aging compared to the nonsolar EV. To quantify the annual effect on transformer aging of SEVs, Fig. 25 shows the

corresponding average annual aging factors. For moderate and high EV penetration rates, which can occur even today in neighborhood clusters, the reduction in transformer aging from using SEVs is significant. For example, at 80% EV penetration, the proposed SEV concept would reduce transformer aging by around 50% in either city compared to the nonsolar EV case, effectively doubling the lifetime of distribution transformers. Specifically, at 80% EV penetration, the LA annual F_{EQA} would reduce from 3.6 to 1.8 (with standard deviation across the 15 transformers equal to 1.1), and the Detroit annual F_{EQA} would reduce from 2.2 to 1.0 (with standard deviation across the 15 transformers equal to 0.54).

While the preceding analysis assumes that drivers charge their EVs immediately on arriving home from their last trip of the day, it is well known that delaying EV charging events can reduce stress on distribution transformers by spreading out the EV load away from the hours of the peak household load [35], [8], [9], [11], [71]. Many current EVs provide a CBD charging option which allows drivers to set a standard daily departure time, and the vehicle calculates the overnight time that the vehicle must start charging to be fully charged by the programmed next-day departure time. This charging start time will vary day to day and vehicle to vehicle, depending on the amount of energy needed by each vehicle on each night. The CBD strategy reduces transformer aging the most out of the currently available in-vehicle options [35], without the need for grid sensors, smart agents, communication networks, connected, and controllable EV chargers, or a method to obtain proprietary vehicle battery SOC data from a variety of EV models. Thus, the CBD charging strategy is studied here to analyze its effect compared to, and combined with, SEVs.

Fig. 26 shows the household and EV charging load for 150 EVs and 150 homes using the CBD charging strategy on a Wednesday night in July. The amount of charging energy reduction from SEVs is the same as with the previous CRA approach, but the EV charging has been shifted to occur

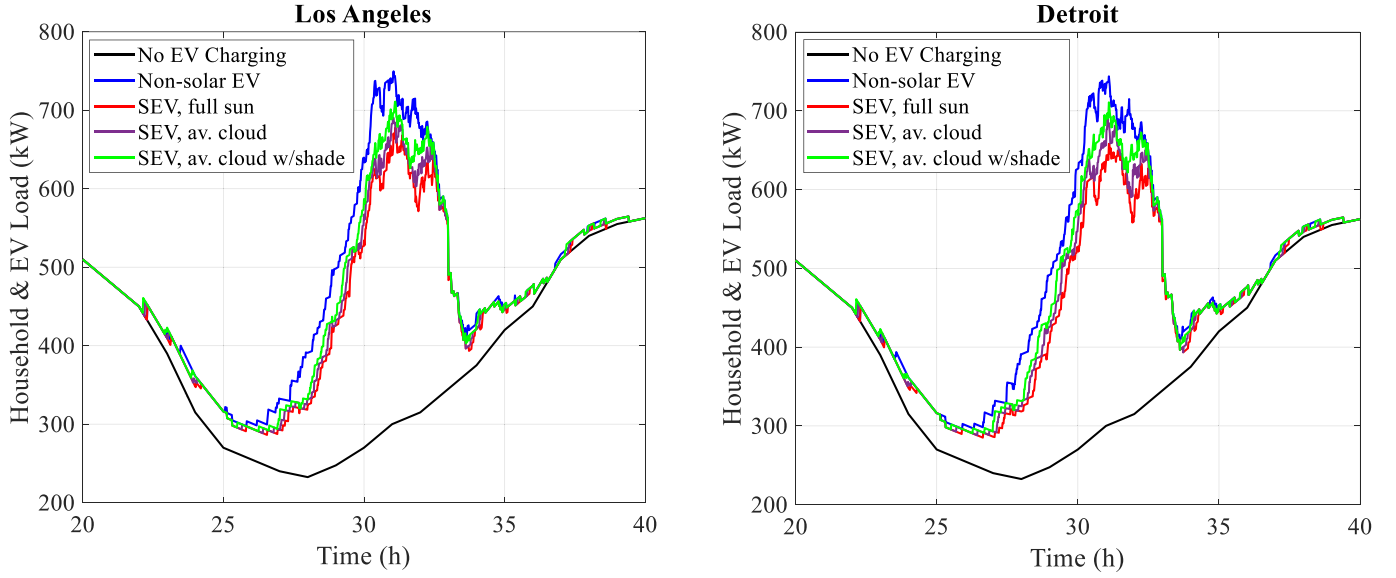


Fig. 26. Household and EV charging load for 150 EVs and 150 houses using CBD charging strategy.

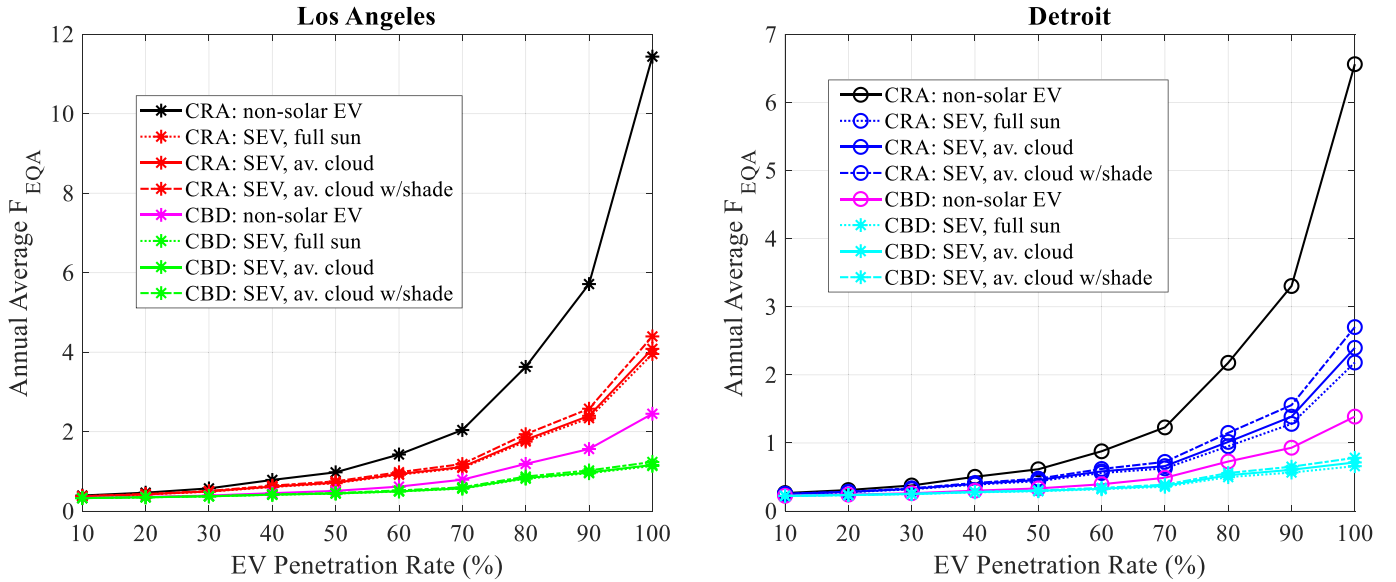


Fig. 27. Average annual equivalent aging factors for nonsolar EVs and SEVs for CRA and CBD EV charging strategies.

mainly in the overnight and early morning. This shift has a large beneficial effect on the associated transformer aging as the EV load is more spread out, has a lower peak, and occurs during the cooler overnight and early morning hours. The resulting annual average F_{EQA} is shown in Fig. 27. CBD charging with nonsolar EVs reduces transformer aging slightly more than the use of SEVs with CRA charging; for example, in LA at 80% EV penetration, $F_{EQA} = 3.6$ for CRA with nonsolar EVs, $F_{EQA} = 1.8$ for CRA with SEVs, and $F_{EQA} = 1.2$ for CBD with nonsolar EVs. The best case is to combine SEVs with CBD charging, which results in average $F_{EQA} = 0.83$ (with standard deviation 0.61 across the 15 transformers). In Detroit, SEVs combined with CBD charging reduced F_{EQA} to 0.53 (with standard deviation 0.38) at 80% EV penetration. While charging strategy selection is

the personal choice of each driver and is thus difficult to enforce, it is clear that SEVs can make a significant reduction in transformer aging with either charging strategy.

C. Driver Benefits

The main driver benefits of SEVs are range extension and reduced charging costs. Both of these benefits stem from an increase in vehicle efficiency in terms of driving distance possible per unit of grid energy charged from the plug. EV energy consumption is commonly expressed in terms of kWh/100 km. In the case of SEVs, the solar energy gained over a day can be considered “free,” and thus, a more useful measure of vehicle energy consumption is “grid kWh/100 km.” Fig. 28 shows the grid kWh/100 km results for nonsolar and SEVs in LA and Detroit, where the results are calculated using

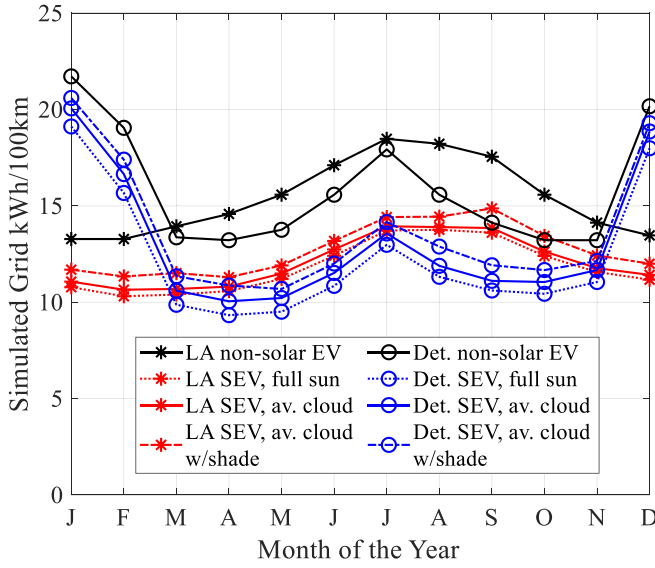


Fig. 28. Simulated average grid energy consumption in grid kWh/100 km in LA and Detroit.

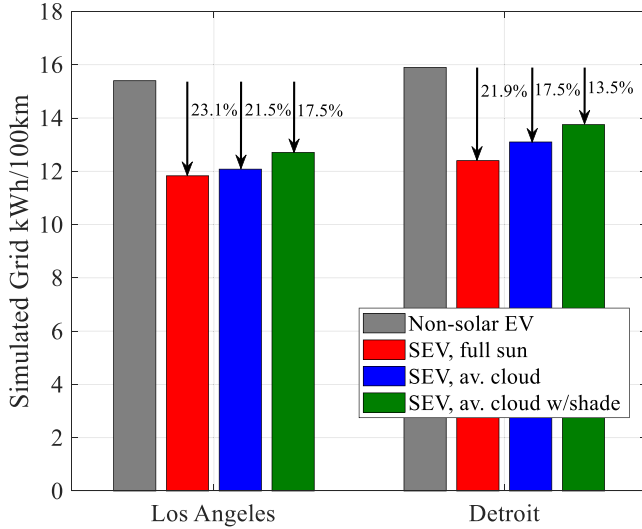


Fig. 29. Simulated annual average grid energy consumption in grid kWh/100 km.

each driver's particular logged drive cycle. This means that these results account for the fact that if a driver does not leave home until late in the morning or afternoon, their vehicle will have been unable to absorb solar charging energy until it begins driving. (Since in this article, each vehicle is assumed to be fully charged from the grid overnight.)

Fig. 28 shows that in both cities, vehicle energy consumption is high in the summer months due to higher HVAC loads. Fortunately, the most solar energy is also available during these months, meaning that the use of SEVs has a significant impact on these months. Fig. 29 shows the annual average grid kWh/100 km. The percentage charging reduction is, as expected, the same as the grid generated energy reductions in Fig. 23. The interesting perspective for the driver is to benefit from essentially a more efficient vehicle. For example, in LA, the annual energy consumption drops from

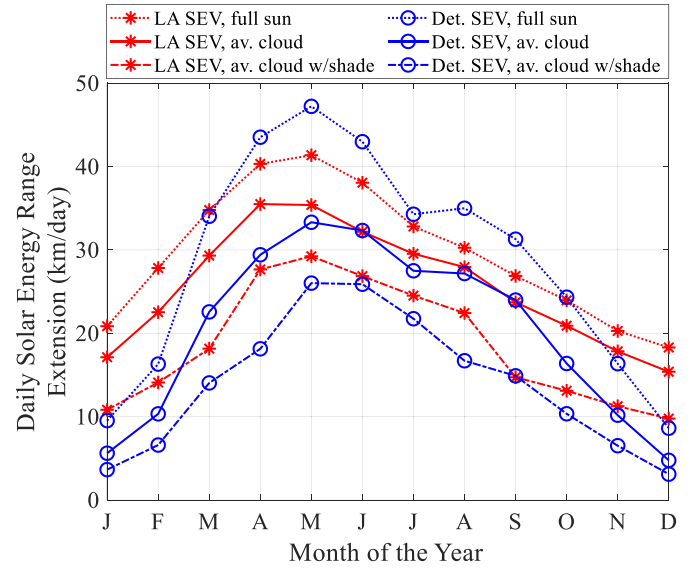


Fig. 30. Daily solar energy range extension in LA and Detroit.

15.4 kWh/100 km for a nonsolar EV to 12.7 kWh/100 km for an SEV with average cloud conditions with partial shading, or equivalently, fuel economy rises from 136 to 165 MPGe.

The gained solar energy allows for range extension during the day's drive, given the fixed amount of grid energy from the plug from the previous overnight charging session. For the range extension analysis only, it is assumed that the driver leaves in the morning so that full solar energy can be captured during the day, in order to find the upper bound of range extension in each month. Range extension will vary by month because of varying solar radiation and also because the vehicle energy consumption varies by month due to varying HVAC loads. Thus, daily solar range extension is calculated per month using (16), where the units are shown in square brackets. $FC_{\text{nonsolarEV}}$ is the fuel consumption of the nonsolar EV as shown in Fig. 28, and E_{solar} is the total daily solar energy available to the EV battery each day, after considering temperature-dependent PV cell efficiency (20% nominal) from Fig. 12 and the 94% power electronic converter efficiency. The average daily per-driver solar range extension results are shown in Fig. 30

$$\text{Solar Range Extension} \left[\frac{\text{km}}{\text{day}} \right] = \frac{E_{\text{solar}} \left[\frac{\text{kWh}}{\text{day}} \right]}{FC_{\text{nonsolarEV}} \left[\frac{\text{kWh}}{100 \text{ km}} \right]} \times 100. \quad (16)$$

The solar range extension in LA is the highest in April, May, and June because a significant amount of solar radiation is available, and the base EV energy consumption is low since air conditioning use is not yet high. Although more solar radiation is available in July and August, the solar range extension is lower because the vehicle is using more energy due to higher air conditioning use. In average cloud conditions, all the months of the year give a solar range extension of at least 15 km in LA. Interestingly, the maximum solar range extension of either city occurs in Detroit in May, and the extensions in April and June are also higher than even the

TABLE I
ELECTRICITY PRICE RATE IN LA AND DETROIT [73], [73]

Month	Los Angeles Consumption Charge (USD/kWh)			Detroit Consumption Charge (USD/kWh)	
	Base M-F 8pm-10am, Sat, Sun	Low-peak M-F 10am-1pm & 5pm-8pm	High-peak M-F 1pm-5pm	Off-peak M-F 7pm-7am, Sat, Sun	On-peak M-F 7am-7pm
January - March	0.17051	0.17405	0.17405	0.05818	0.27710
April - May	0.17463	0.17817	0.17817	0.05818	0.27710
June	0.17073	0.19446	0.25657	0.05818	0.27710
July - September	0.17542	0.20286	0.26126	0.05818	0.27710
October - December	0.16788	0.17142	0.17142	0.05818	0.27710

peak solar range extension in LA. The reason is that in the spring months, Detroit has high solar radiation but moderate temperatures, meaning that HVAC use is low or zero. Since the vehicle is not only obtaining a large amount of extra solar energy but also driving in a very efficient way due to low HVAC use, the solar range extension is the highest at this time. In average cloud conditions, 8 months of the year give a solar range extension of at least 15 km in Detroit. On an annual average, the daily solar range extensions for LA are 30 km for full sun, 26 km for average cloud conditions, and 19 km for average cloud conditions with partial shading. On an annual average, the daily solar range extensions for Detroit are 29 km for full sun, 20 km for average cloud conditions, and 14 km for average cloud conditions with partial shading.

The second driver benefit of SEVs is charging cost reduction. Time-of-use (TOU) electricity billing has been assumed for all drivers in each state. In LA, there are three TOU billing rates: base, low peak, and high peak. In Detroit, there are two TOU billing rates known as OFF-peak and ON-peak. Table I shows the TOU billing rates across all months in both LA and Detroit [12] [72].

EV charging cost is calculated using the following equation:

$$\text{Charging Cost} = E_{\text{TOU1}} C_{\text{TOU1}} + E_{\text{TOU2}} C_{\text{TOU2}} + E_{\text{TOU3}} C_{\text{TOU3}} \quad (17)$$

where E_{TOU1} is the energy consumed in kWh by the charging vehicles in the first TOU category (i.e., base rate for LA), C_{TOU1} is the \$/kWh cost for the first TOU category as per Table I, and subscripts 1, 2, and 3 refer to each TOU category (with only two categories used for Detroit). Taxes of the respective cities are then added to this consumption charge (10% for LA [73] and 4% for Detroit [74]). Fig. 31 shows the monthly EV charging cost savings per driver for using SEVs in place of nonsolar EVs. The summer months provide the highest cost savings for SEVs because there is higher solar incident energy. For days with average cloud conditions, the annual projected charging cost savings are \$182 in LA and \$73 in Detroit, which are 21.6% and 19.3% of total EV charging costs, respectively, as shown in Fig. 32.

D. Environmental Benefits

The main environmental benefit of widespread SEV use is a reduction in electricity-related emissions due to the displacement of grid charging with onboard solar charging. This analysis quantifies the reduction in CO₂ emissions specifically

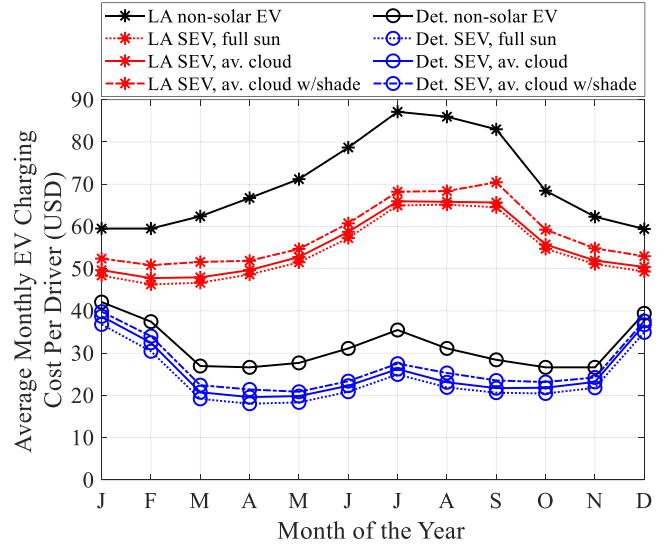


Fig. 31. SEV effect on monthly EV charging costs per driver in LA and Detroit.

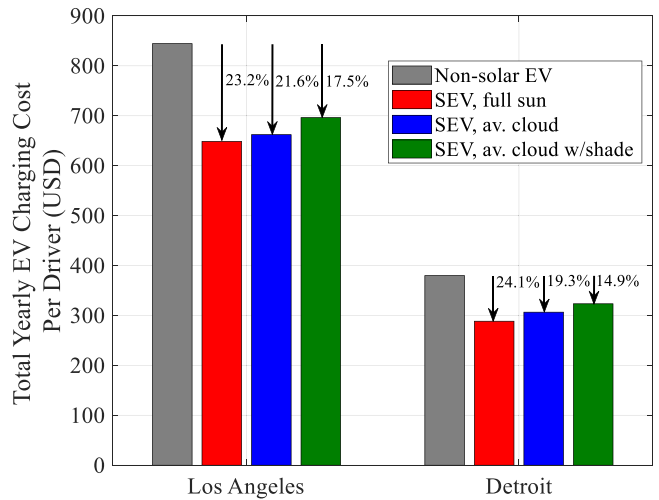
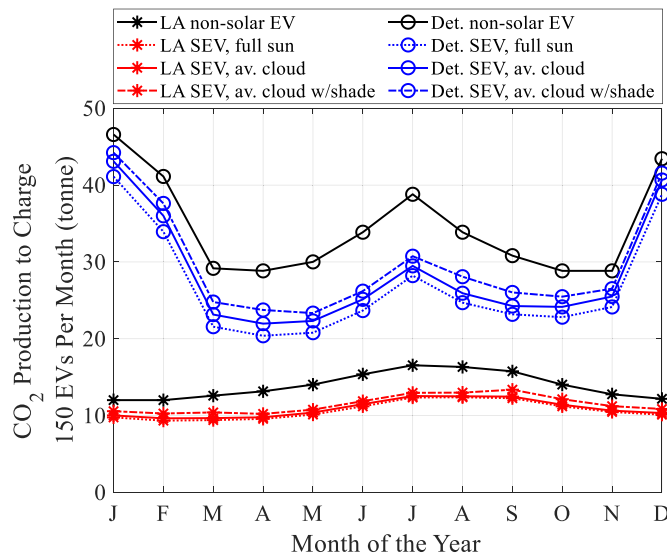
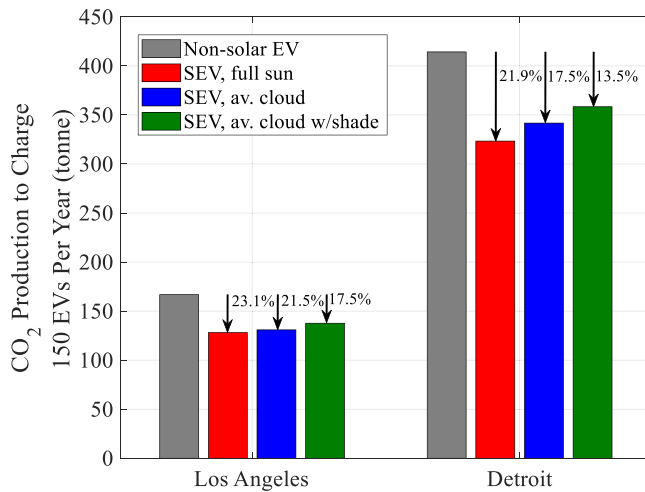


Fig. 32. SEV effect on annual EV charging costs per driver.

as it is a prevalent greenhouse gas. The carbon intensity for the current grid mix in California is 215 kg/MWh, and the carbon intensity in Michigan is currently 520 kg/MWh [75]. California has a much lower carbon intensity than Michigan because its grid mix consists of 37% natural gas and 63% nonemitting sources (nuclear, hydroelectric, solar, and wind),

Fig. 33. CO₂ production to charge 150 EVs monthly in LA and Detroit.Fig. 34. Annual CO₂ production to charge 150 EVs.

whereas the Michigan grid is powered by 37% coal and 27% natural gas, with nonemitting sources providing the other 36% [76]. Fig. 33 shows the amount of CO₂ produced monthly to meet the charging demand of 150 EVs for the nonsolar EV and SEV cases in LA and Detroit. Fig. 34 shows the comparison of annual CO₂ production in both the cities to charge 150 nonsolar EVs and 150 SEVs in case of full sun, average cloud conditions, and average cloud conditions with partial shading. Annually, the use of SEVs instead of nonsolar EVs would reduce CO₂ emissions approximately by 239 kg per driver in California (21.5%) and 484 kg per driver in Detroit (17.5%) considering average cloud conditions throughout the whole year. California currently has about 500 000 EVs on the road [77]. If SEV emission reductions are projected across this fleet, the annual CO₂ reduction for this state is 119 500 tonnes.

E. Economic Analysis

The cost of PV integration is a crucial factor in the viability of the SEV concept. Utility-scale PV costs are useful for

these estimates, where silicon module costs have fallen to <\$0.25/W [78] with installed costs <\$0.89/W [79]. Cell and module costs have fallen for decades due to the experience curve, and this trend is expected to continue for some time. Full integration of the cell manufacture brings the installation costs into a highly automated vehicle manufacturing environment, with opportunities for significant cost reduction as compared to highly labor-intensive PV installation. Furthermore, the energy-intensive racking components for PV installation are not required. From this point of view, the manufactured cost of PV for SEVs could be reduced well below the installed cost for utility-scale PV. This article conservatively estimates the PV cell and integration costs to be \$1.00/W, based on an estimated CIGS cell cost of \$0.50/W [80] and an additional \$0.50/W for the PV integration with the automotive steel, totaling \$620 for a 620-W peak power system. While this delineation is artificial in an integrated process, the latter cost recognizes the process development and infrastructure updates with respect to the conventional automotive manufacturing process. For this SEV application, the PV balance of system costs is from the DC/DC converters required to control and step up the solar power. The DOE has set a target of \$50/kW in 2020 for automotive-grade DC/DC converters [21]. Thus, this article estimates the cost of a 620-W DC/DC converter at \$31 plus \$19 for wiring, for a total of \$50. This brings the total mass-produced cost premium for an SEV to \$670, which is on par or less than many other options offered on new vehicles. These rough cost estimates are predicated on achieving the full benefits of a fully integrated process implemented at scale.

By considering only the driver's charging costs, the payback period for the driver assuming a \$670 premium for the SEV is 3.7 years in LA and 9.2 years in Detroit. However, these metrics do not fully capture the other SEV benefits, such as driving range extension, reduction in CO₂ emissions, and benefits to the grid. This fact highlights one current difficulty in moving toward mass-produced SEVs: the benefits are widespread across numerous players, so motivation for investment by single parties, such as vehicle manufacturers, is not yet high. This is analogous to the high current payback periods for drivers considering EVs compared to ICE vehicles—the environmental benefits add additional purchase incentive beyond traditional payback periods.

V. SEV FUTURE TRENDS

This article has focused on analyzing system-level SEV benefits for a typical passenger vehicle under the standard one-driver model. Future SEV applications could also include autonomous and ride-sharing EVs. For a fully autonomous EV, the vehicle auxiliary electrical load is expected to increase by a few kilowatts for powering sensors and additional onboard computers [81]. This additional load would increase EV charging power needs and reduce the driving range. For a one-driver autonomous vehicle, the low-cost integrated PV cells are an attractive option because the harvested solar power could help offset the increased energy consumption of the vehicle compared to a baseline EV. Ride sharing of a nonautonomous vehicle would change the drive cycle dramatically such that the vehicle would likely drive for

much more time in the day compared to the one-driver data used in this analysis. This could increase the vehicle energy expenditure significantly over a 24-h period, with a potential for near-constant driving conditions. For a ride-sharing SEV, even though the solar energy captured would be equal to that in a one-driver SEV, the percentage of SEV energy use reduction (and the associated benefits) would be smaller due to the larger energy expenditure. Overall, autonomous vehicles and ride-sharing models increase vehicle energy use—SEVs can play an important role in partially mitigating these increases though the percentage of SEV benefits will reduce with larger and larger vehicle energy consumption requirements.

There has been recent industry interest in using thin-film solar panels on Class 8 tractors and trailers [82]. For both diesel and electric tractors, the energy-saving percentage from using onboard solar is small due to the large size and mass of the truck, and the fact that it can drive for the majority of the day, leading to enormous daily energy needs. However, onboard solar can help with idle reduction and jump-start reduction (from drained batteries) by powering electrical loads such as trailer refrigerators, lift gates, and HVAC. This use of onboard solar with diesel trucks can increase driver convenience and extend battery life.

SEVs also present the opportunity to use EVs as decentralized solar generation systems if a bidirectional onboard charger is included in the vehicle. Much prior work has focused on the design and control of bidirectional onboard chargers for EVs for the purpose of providing power from the EV battery to the grid for V2G functionality [83]–[86], but no work has considered its use in solar-charged EVs. Bidirectional chargers are more expensive than unidirectional chargers because switches must be used in place of the AC/DC rectifying diodes. A further difficulty is that the control complexity is increased. For the proposed SEV concept with a bidirectional onboard charger, vehicles fully charged overnight and at home for part or all of the following day could power the house, reducing electricity costs and acting as a decentralized solar generation system with built-in energy storage. The benefit would depend on the amount of time that the vehicle is parked at home while fully charged, which would change driver to driver.

To quantify the potential benefit, an additional analysis was run on the 150 unique drivers using a 94% efficient bidirectional onboard charger, where a fully charged SEV provides power to the house before the first departure of the day. In this case, the annual projected net charging cost savings are \$218 in LA and \$99 in Detroit, which are 25.8% and 26.0%, respectively, of the total charging costs. Although the amount of energy gained by the home is relatively small, the high TOU electricity prices in the daytime mean that a reasonable benefit is achieved, on average. Of course, for drivers who regularly leave early in the morning, the benefit of bidirectional power flow would be negligible, and for drivers who regularly leave later in the day, the benefit would be significant. An error analysis shows that across 150 drivers, the standard deviation of total charging cost savings in LA is \$56.50 and, in Detroit, \$37.60 with the availability of SEV2G feature. To analyze the economic feasibility, the approximate

cost increase of a bidirectional onboard charger compared to a unidirectional onboard charger is calculated as follows: Infineon switches IPW65R080CFDA (\$10.78 each) are assumed to replace eight diodes (IDFW40E65D1E at \$5.55 each) in an onboard charger, 100% of the switch cost is added for gate drivers and thermal management of the switches, and the total cost difference is halved to account for the large purchasing volumes of the automotive sector. The resulting hardware cost increase is \$64 per charger, meaning that the average payback period in LA is 3.5 months and, in Detroit, 7.8 months, for the bidirectional charger. This does not account for the extra initial development costs of the bidirectional onboard charger. Overall, this analysis shows that adding a bidirectional charger to an SEV to further reduce net charging costs reduces the total payback period (3.4 years in LA and 7.4 years in Detroit).

When considering the environmental benefits, a bidirectional solar power V2G feature would reduce the total annual CO₂ emissions by 290 kg per driver in LA (standard deviation 70.6 kg across 150 drivers) and 563 kg per driver in Detroit (standard deviation 129.5 kg across 150 drivers), compared to nonsolar EVs. This V2G concept can be further extended to a smarter SEV2G energy management system. With the usual V2G framework, a vehicle discharges its battery to provide services to the grid—this involves a trade-off between battery energy needed by the driver and the value of the energy provided to the grid. For an SEV, this trade-off changes because the vehicle is a mobile solar energy generating station and thus could be more able to exchange power with the grid with minimal impact to the driver. Thus, future research on reducing the cost and complexity of bidirectional onboard chargers would be valuable to SEVs.

Although the fully integrated SEV concept proposed here has been shown to have promising benefits on a system level, there are numerous practical challenges to be considered in the future. First, one concern is how to deal with automobile accidents which damage one or more of the solar body panels. For damaged body panels, it is currently common for whole panels to be replaced with new panels from the vehicle manufacturer. For the SEV case, the difference would be that the new panel with integrated solar cells would be slightly more expensive than the nonsolar panel. However, if the mass production of solar cells integrated onto the steel is able to reduce costs to a reasonable level for whole vehicle purchase, then the cost of a solar replacement panel could similarly be in a reasonable range.

Second, there will be a challenge of developing a marketing strategy for SEVs. The environmental-friendly aspect of SEVs will be important, just as people buy EVs today that cost more than their ICE counterparts, often for environmental reasons. The esthetics of an SEV may also be a marketing challenge, but there is a potential to turn it around as a positive point if the SEV can be made a desirable status symbol—this is similar to the Tesla EV marketing strategy. Furthermore, SEVs will be less suitable and less popular for drivers that routinely park indoors during the day and may be less desirable to drivers living in more northern and/or cloudy cities—this will affect the marketing plan and business model.

Third, since the proposed SEV concept includes many critical parts from different fields, it is important that the cross-functional team working on its production perform a failure mode and effects analysis (FMEA) to identify potential failures in a wide range of areas, including solar-steel integration, PV durability in harsh weather and over time, and power electronic converter reliability. Finally, a major challenge is an investment needed to develop the process to integrate the PV directly onto the steel, including smooth adhesion over curved surfaces and durability concerns over many years of use in potentially harsh weather. As described in this article, many parties can benefit from SEVs, but one party must ultimately become motivated enough to invest in this integration process.

VI. CONCLUSION

The big-photograph benefits of SEVs have not yet been realized because the thinking has been small (vehicle rooftop panels) and limited in scope (considering only driver benefit). This article has modeled the effect of panel tilt, vehicle direction, and partial shading on solar generation and has shown that the proposed low-cost SEV concept offers real benefits to the grid, driver, and environment, indicating that further research is warranted to address the design and manufacturing challenges of SEVs. For the given data set of 150 drivers over 1 week, the use of SEVs (compared to nonsolar EVs) would reduce the annual net vehicle energy consumption by 21.5% in LA and 17.5% in Detroit under average cloud conditions including temperature-dependent PV efficiency. The peak solar range extension is 47 km/day in Detroit on a sunny day in May, and the charging costs are reduced by about 20% for average cloud conditions. Although each benefit is useful alone, the real potential of SEVs is clear when considering the combination of these results together. Thus, a low-cost SEV has the potential to empower a massive transition to solar power on a distributed scale of individual car owners.

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